

flea

NEWS 94 *October 2024*



FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Before including in publications any of the citations given here, please review for accuracy and correct formatting. Many of our sources are abstracting journals and current literature sources, and editors have not necessarily checked citations.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Citations for new flea species descriptions, Flea research citations from journals or conference proceedings that are not available or indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Editorial

"Flea pupae"

"The pseudosetae have not yet been studied in the pupal stages but Jordan... drew attention to the fact that if the mesonotum is injured in larval life the pseudosetae may develop into spines similar to those in the pronotal comb".

- Rothschild (1975)

"I believed that this [spandrel] concept—a predictable form that arises as a side consequence rather than a direct adaptation—had important application to biology but lacked a name. (Darwin's "correlation of growth" came to mind, but his term is ambiguous and never caught on in any case.) I thought, and continue to feel, that spandrel is the most obvious, the most useful, and the most historically sanctioned term available for such a central concept. Moreover, the term spandrel is particularly applicable to biological problems of adaptation because the architectural concept stresses the same point of distinction between historical origin and later utility that has proven so troublesome in evolutionary theory."

- Gould (1997)

Dear *Flea News* Reader,

In the last of the series on immature fleas, I discuss the least known, but quite interesting stage, the flea pupa.

The pupa may show traits that are not necessarily subject to direct selection because the pupa is inactive in a cocoon. However, these traits are necessary byproducts developmentally that may form important adaptive traits in subsequent stages. Such traits have been termed "spandrels" (Gould and Lewontin 1979). These characteristics make immature fleas particularly valuable to understand "evo-devo" (evolution and developmental biology).

Does the flea show spandrels?

Studying the pupa is facilitated with the "naked pupae" technique (Dryden & Smith 1994, Lawrence & Foil 2002): Wandering larvae, about to spin cocoons, are placed in round-bottom microtiter wells. Such containers seem to prevent the larvae from gaining purchase to spin a cocoon, so the larvae pupate without a cocoon. Naked pupae allow easy viewing of exterior anatomy.

Effects of temperature, humidity, infection (cestodes), and insecticides on pupae of limited flea taxa, mainly Pulicidae, have been examined (Mellanby 1933, Snodgrass 1946, Sharif 1949, Marshall 1967, Margalit & Shulov 1972, Cerqueira and Linardi 1979, Silverman et al. 1981, Linardi & Guimarães 1993, Metzger & Rust 1997, Miller et al. 1999, Krasnov et al. 2001, Kreppel et al. 2016).

Observations on flea pupae revealed puzzling structures when Sharif (1937) saw what he hypothesized were vestigial wing-buds. However, Poenicke (1969) and Rothschild (1975) disputed this, noting that the structures were in Ceratophyllidae, but not Pulicidae. Other puzzling flea structures that could be spandrels with developmental significance include the frontal and deciduous tubercles found in some fleas (Snodgrass 1946, Medvedev 2015), combs (Medvedev 2017), spiracle and trachea elaborations (Beaucournu and Laudisoit 2009), and unciform sclerotizations ("consistent in a given species, unciform sclerotizations are random with respect to genus or other taxa", Beaucournu et al.

2013).

Pupal homologs and synapomorphies (shared, possibly derived characters that may indicate common ancestors between Mecoptera and Siphonaptera) were proposed by Kluge 2002, who concluded that "The pupal leg buds may replicate the initial position of larval legs in the flea ancestors (whereas in the present-day fleas, the larval legs are completely lost).".

I propose that the flea pupae "wing-buds" are actually spandrels - necessary structures to develop the specialized spiracles, trachea, and musculature of the Ceratophyllidae adult flea. An advanced form of respiratory system may adapt fleas to dry or cold environments, and could have facilitated the diffusion of newer fleas like Ceratophyllidae around the globe.

Flea pupae and the other immature stages are intriguing and under-studied, especially in regard to diapause, developmental temperature, water relations (Fielden et al. 2004), and embryology. Their characteristics are probably as or more important for understanding some aspects of flea life, their ecology and evolution, as is the well-studied adult (Cerqueira and Linardi 1979, Linardi & Guimarães 1993, Szpila et al. 2024).

Yours in fleas,
R.L. Bossard
Editor, Flea News

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Announcements

Lewis World Flea Species List Updated

Available online at Catalogue of Life Checklist, GBIF (Global Biodiversity Information Facility):

Citation: Hastriter, M. W., & Bossard, R. L. (2024). Robert E. Lewis World Species Flea (Siphonaptera) List (version 2.0). In O. Bánki, Y. Roskov, M. Döring, G. Ower, D. R. Hernández Robles, C. A. Plata Corredor, T. Stjernegaard Jeppesen, A. Örn, L. Vandepitte, T. Pape, D. Hobern, S. Garnett, H. Little, R. E. DeWalt, K. Ma, J. Miller, T. Orrell, R. Aalbu, J. Abbott, et al., Catalogue of Life (Version 2024-09-25). Catalogue of Life, Amsterdam, Netherlands. <https://doi.org/10.48580/dfvll-48fy>

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Transfer of Collection

The Ectoparasite Collection, especially that of Siphonaptera, housed at the Department of Parasitology, Federal University of Minas Gerais (UFMG), Brazil, is now deposited at the René Rachou Institute, Oswaldo Cruz Institute Foundation (FIOCRUZ), Avenida Augusto de Lima, 1715, 30.190-002, Belo Horizonte, Minas Gerais State, Brazil.

It was transferred after the retirement of Professor Pedro Marcos Linardi from the Federal University of Minas Gerais, who organized and expanded it for over 55 years of research and scientific activities. The current curator is Dr. Daniel Moreira de Avelar, e-mail: aveldan@gmail.com

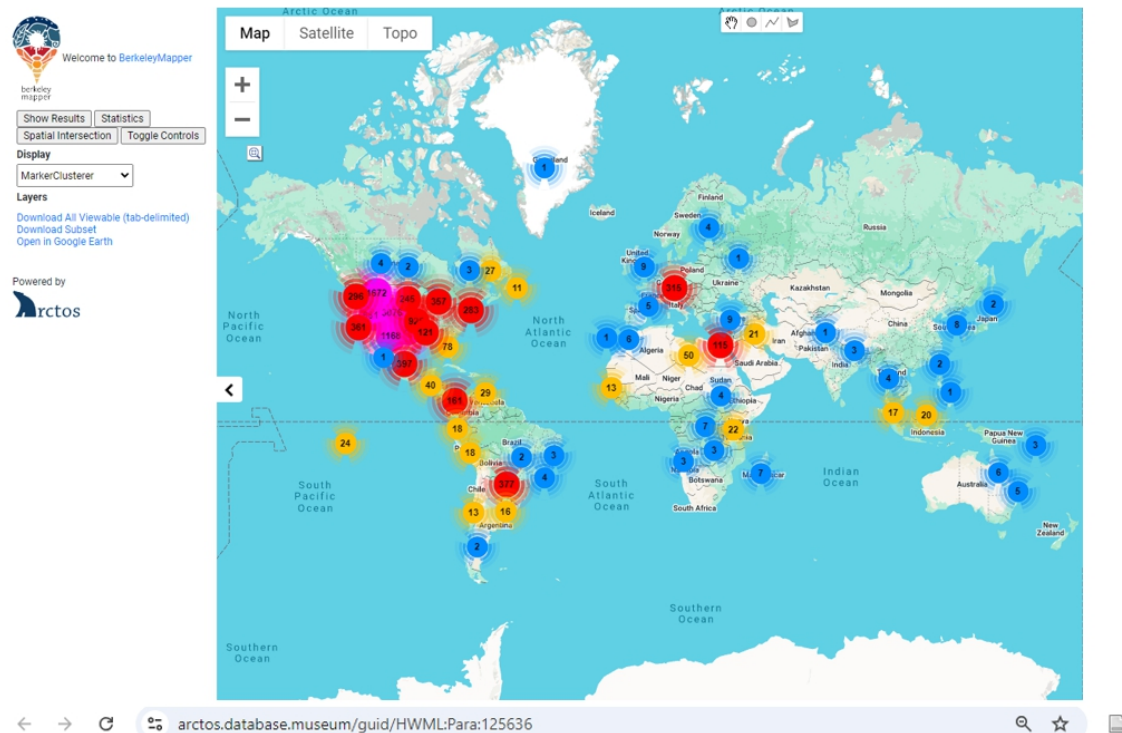
[Editor's note: there are plans to make this collection accessible online.]

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REPORT ON THE HOPLA FLEA COLLECTION

The Hopla Flea collection donated to the Manter Laboratory is now fully entered into our HWML [Harold W. Manter Laboratory of Parasitology, Lincoln, Nebraska, USA] database on the multi-institutional database "*Arctos*". This work is supported by a Grant to the Manter Lab from the NSF (Natural History: Digitizing and conserving specimens in the Manter Laboratory of Parasitology. National Science Foundation. DBI-1756397).

Specimens (collected 1950 through the early 1990s) were georeferenced in the Manter Lab using automated tools to facilitate mapping host and flea species distributions. The DATA include 45,640 records (slides of fleas) with 23,974 images that are in JPG format, 40MP resolution, ~20MB each image.



Manter Parasite Collection ▾
University of Nebraska State Museum ▾

HWML:Para:125636

Myodopsylla collinsi

Parts: whole organism (slide)
specific locality not recorded
United States, Oklahoma, Greer County
18-Mar-1965 (1965-03-18)
Record Type: PreservedSpecimen

Identification

Media

Agents

Attributes

Locality

Identifiers

Parts

Identifications

[1] **Myodopsylla collinsi** 
 ◯ Animalia, Arthropoda, Insecta, Siphonaptera, Ischnopsyllidae, Ischnopsyllinae, Myodopsylla,
 Myodopsylla collinsi

Identified by **Cliff E. Hopla** 1965-03-18

Attribute	Value	Determiner	Method	Date	Remark
nature of identification	features				

Agents

collector

Cliff E. Hopla 

verification agent

Agents acting only in method-specified role. Please use the "comment or report bad data" link if you have more information.

Agent	Method	By	Date	Remark	More
Tom Gray		Cliff E. Hopla 	1965-03-18		search 

Identifiers and Relationships full split view ▼

Identifiers

Identifier	IssuedBy	Type	More	AssignedBy	Remarks
13180365-A		field number	search	Gabor R. Racz  @2022-09-30	
CN5395		collector number	search	Gabor R. Racz  @2022-09-30	

Catalog Record Media

Media linked to this Cataloged Item.

Showing results 1 - 5 of 7

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image (image/jpeg)

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image (image/jpeg)

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image (image/jpeg)

[Media Details](#)



image (image/jpeg)

[Media Details](#)

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 scaling information in JPG c... scaling information in JPG c... scaling information in JPG c... scaling information in JPG c...



Catalog Record/Curatorial

Record Identifier: <https://arctos.database.museum/guid/HWML-Para>

Accession: [View P-2017-068](#)

Example of a flea photo

<https://arctos.database.museum/guid/HWML:Para:125649>

HWML:Para:125649

Lagaropsylla from a bat (Chiroptera) Collected in the Congo (current Democratic Republic of Congo) in the 1950's by Harry Hoogstraal.



SUBMITTED BY SL GARDNER 15 SEPTEMBER, 2024

Scott Lyell Gardner, Ph.D., AAAS Fellow, Curator, and Professor, Harold W. Manter Laboratory of Parasitology W-529 Nebraska Hall University of Nebraska State Museum and School of Biological Sciences, University of Nebraska – Lincoln, Lincoln, Nebraska 68588-0514

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ASP Page: <http://asp.unl.edu>

<https://www.facebook.com/ManterLab>

[Editor's note: "The HWML keeps its database on *Arctos* and *Arctos* feeds our data to GBIF [Global Biodiversity Information Facility]. So all of the data are available on GBIF after we enter it onto *Arctos*."]

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Presentation at 52ND ANNUAL CONFERENCE SOCIETY FOR VECTOR ECOLOGY

FORT COLLINS, COLORADO SEPTEMBER 15 – 19, 2024

Expanding the toolbox for flea control and plague mitigation in rodent populations.

David Eads deads@usgs.gov U.S. Geological Survey, Fort Collins, CO.

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Seminars of Interest to Parasitologists

Free Online

1. Ecological Radiations of Mammals: Before & After the K/Pg Mass Extinction

Gregory Wilson Mantilla, University of Washington, Burke Museum of Natural History

<https://doi.org/10.52843/cassyni.xg75qh>

2. Extinctions along the last legs of the first human migrations to Australia

Prof. Julien Louys, Griffith University

<https://doi.org/10.52843/cassyni.k9y1yq>

3. Conservation Biogeography of Australasian Marsupials

Assoc. Prof Diana Fisher, University of Queensland

<https://doi.org/10.52843/cassyni.4j8sdx>

Abstract: "The diverse and distinctive Australasian marsupial fauna has experienced considerable loss over the last 200 years. Seventeen of the 176 Australian marsupial species have become extinct in this period (~10%), far higher than the global average (1.4%). Many extinct Australian marsupials were formerly abundant and widespread. About 40% of Australian marsupials are now threatened. The pattern of decline in Australian marsupials is broadly similar to that of Australia's native rodents- invasive mammalian predators and land use that facilitates them are the major problems. Knowledge of the marsupial faunas of Papua New Guinea and Indonesia is relatively poor. Both regions have high endemism. There are 96 species of marsupials on the main island of New Guinea and surrounding small islands, and 15 species in Wallacea (the islands of Indonesia and East Timor east of Wallace's line) including five shared with New Guinea. Although none are listed as Extinct, two Critically Endangered marsupials are flagged "possibly extinct." Most threatened New Guinean and Wallacean species follow the globally typical pattern of high vulnerability in island endemics and small-range mammals with specialist habitats. The major threats to marsupials of New Guinea and Indonesia are habitat loss, hunting, and low levels of awareness and investment in

conservation and research. Many Australasian marsupial species – particularly mountain-top endemics are increasingly threatened by climate change."

These seminars introduce three of the contemporary researchers on mammal evolution.

Radiations of mammals, the subject of the first seminar speaker Professor Mantilla, has been a mystery. The origin of the placenta is unclear. Insectivores are now a polyphyletic group. Marsupials never seemed common in Africa, so what was there? New fossils and understanding of the timing, origins, dispersal, and extinctions of mammalian lineages are critical to understanding the diversity of fleas and other parasites.

The second seminar looks at fossil discoveries by Professor Louys of Holocene (since approximately 11,700 years ago) animal extinctions in the Indo-Malay biogeographical realm.

This realm is poorly known biotically and variable in extent, depending on the biota examined, usually including India, SE Asia including Indochina and southern China, Malaysia, and western Indonesia. It is demarcated by various hypothetical biotic boundaries such as the Kangar-Pattani Floristic Line, Lydekker Line, Wallace Line, and Weber Line. Eastern Indonesia (mainly Papua New Guinea) is generally considered in the Australian realm.

The Indo-Malay area looked much different in the Pleistocene (2.58 million to 11,700 years ago) because of glaciation with the sea level lower and many of the islands we see today in the 17,000 islands of Indonesia connected by an terrestrial plain called Sunda Land (Sunda Shelf) that is now underwater, much like the Beringia and Doggerland, southern North Sea of today, were once dry land.

The seminar speaker Louys expounds further how migration of *Homo sapiens* into Indo-Malay had several coincidental events. Hominids disappeared such as *Homo erectus* and *Homo floresiensis* (the "Hobbits"). Pongids (orangutans) were formerly widely distributed in open savannah, but are largely extirpated except for small areas of deep forest.

This seminar notes that the megafauna, such as species related to elephants and giant lizards, seem to disappear with the appearance of humans approximately 10,000 to 50,000 years ago, consistent with data from throughout the globe. Not until much later (a mere 1,500 years before the present day) did smaller species in Indo-Malay such as giant rats (*Comphymys*, and *Ratini*) that are several times larger than the extant black rats, experience extinctions coincident with the appearance of metal tools and widespread deforestation by humans of trees such as sandalwood. This suggests to Louys that human technology can either diminish or preserve biodiversity, depending on how it is used.

Undoubtedly, these giant rats hosted fleas, so like the Neotropical realm with loss of marsupials, the Indo-Malay realm placental mammals probably experienced extinctions of their parasites.

The last seminar introduces some of the astonishing and beautiful diversity of marsupials and monotremes in Australian realm. These animals can be grouped according to the presenter Professor Fisher into two groups: arid land animals inhabiting the western 2/3 of Australia, and animals of the wet forest inhabiting the eastern third, plus islands to the north such as New Guinea.

Surprisingly, it is the arid land marsupials that are most endangered because of land use and invasive eutherian (placental) mammals such as fox, rabbit, cattle, sheep. This is not to say all the marsupials in the wet forest are doing okay; they are not. Many are extinct or critically endangered.

What does this have to do with fleas? Marsupials are host to interesting parasites. Many fleas that are considered primitive or basal to other fleas are on these mammals, including Pygiopsyllomorpha, *Macropsylla*, Stephanocircidae (helmet fleas), and Doratopsyllinae (marsupial and shrew fleas). This last group is represented in Australia by two genera: *Acedestia* and *Idilla*.

I wondered why these two genera? After viewing this seminar, I surmised that *Acedestia* seems to be associated with the arid land ecosystems of the south and west Australia. and *Idilla* seems to be in wet forests of east Australia. These fleas are likely adapted to contrasting thermal moisture regimes, as we may see with the deer mice fleas in North America, *Aetheca wagneri* in the west and *Orchopeas leucopus* in the east.

These seminars are worth watching if you are interested in mammal evolution.

R.L. Bossard

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Giant pangolin found in Senegal

Ndiaye, M. M., Drouilly, M., Senghor, A. A., Thiaw, O., Ingram, D. J., Shirley, M. H., Strampelli, P., Marti, C. M., & Henschel, P. (2024). Rediscovery of the endangered giant pangolin (*Smutsia gigantea*) in Senegal after 24 years. *African Journal of Ecology*, 62, e13279. <https://doi.org/10.1111/aje.13279>

Pangolins have generated great interest in recent years, largely due to the unprecedented scale of trafficking the species experiences. Detailed knowledge of pangolin distribution in West Africa is lacking, but many local extinctions are suspected. The last documented sighting of a giant pangolin in Senegal was in Niokolo-Koba National Park (NKNP) and dates back to 1999. Following a large camera-trap survey carried out in NKNP in 2023, we present photographic evidence that giant pangolins are still present in the park. Such rediscoveries not only underscore the importance of systematic biodiversity inventories, but also the critical value of West Africa's large protected areas.

Zhi-Cheng Liu, Peng Cen, Jian-Kun Sun, Fu-Hua Zhang, Amna Mahmood, Chen Lei, Shi-Bao Wu. 2024. Chinese pangolins (*Manis pentadactyla*) are not functionally extinct in China. *Zoological Research: Diversity and Conservation* 1(1): 79-81. DOI: 10.24272/j.issn.2097-3772.2023.297

"Life History Strategies Among the Siphonaptera"

by A.H. Benton

Transcribed and annotated by R.L. Bossard

[The following are notes of A.H. Benton, unpublished to my knowledge, that was as I recall the basis for a speech by him at the University of Utah circa 1984. C.E. Hopla commented on the term "strategies", do fleas really "strategize"? Many of the questions remain unanswered. R.L. Bossard]

Many years ago, I used to teach a course called field biology, and one of the requirements of the course was the performance of a small research project. Since it was a two hour one semester course, the project was of necessity pretty minor, and a good percentage of the students failed to secure any significant results. In those cases their paper at the end of the semester usually closed with some variation on this sentence: "I never realized how hard it is to establish a simple fact".

This confession is the beginning of scientific wisdom. I always felt that those who wrote such sentences had achieved a large part of what I had hoped they would achieve in doing the project. What I am going to tell you today is what I believe to be some facts about flea life histories. You will notice many gaps, many "maybes" and "probablys" and "I thinks". After all I've been working on fleas only about 35 years, so there's a lot I don't know yet.

I would like to begin by reviewing the flea life cycle, in case you have forgotten. Fleas are, of course, holometabolous insects, which means that the adults lay eggs, small oval whitish structures which hatch out into maggot-like larvae. These larvae feed voraciously on bits of skin and feces and blood and on each other, and no doubt on other things we don't know about [1]. They undergo three instars [2], and then pupate. The pupa is usually contained within a silken cocoon, to which may or may not be appended bits of sand or other detritus [3]. This pupa undergoes the necessary transformations, and emerges in due time as the adult flea. Adult fleas may live for a year or more, but probably is only a month or so during which time they feed, mate and produce the next generation. Most fleas move on and off their host quite freely, and mating and oviposition take place mostly in the nest or burrow or other habitat of the host rather than on the host itself [4]. If oviposition occurs on the host, most species produce eggs which will simply fall off and undergo development wherever they fall [5]. Larvae are very rarely found on the host, but it may be normal for larvae in severe arctic regions to stay on the host, at least in some species [6].

Now on to life history strategies. Obviously, this term covers a multitude of sins. It can refer to host selection, to reproductive strategy, to larval habits, to emergence, to host finding, and so on. Let's begin with some variation on the typical life cycle and habits of fleas.

Within one family, the Tungidae, the adult flea attaches itself to the host in a rather permanent manner, the female burrowing into the skin of the host and then swelling to a relatively enormous size. The process is peculiar in that growth and change occur in the imago stage, and this is called neosomy. The head and thorax usually remain relatively unchanged, but the abdomen swells from 100 to 1000 times and the female quickly produces eggs. In one species, the males do not feed as larvae, but pupate

on the host, emerging after the female has engorged. They mate and the male dies, without ever having fed, having lived its entire lifespan with only the energy contained in the egg [7].

The American sticktight flea (*Echidnaphaga gallinacea*) attached firmly and is partially imbedded [sic, embedded], but does not undergo neosomy. This is our smallest pulicid, and one of our most eclectic feeders: common on poultry, dogs and many wild mammals. It is one of the very few fleas to feed regularly on both birds and mammals.

Within the Vermipsyllidae, mostly parasites of carnivores and ungulates, neosomy also occurs [8]. One Asiatic species, *Vermipsylla alakurt*, which feeds on sheep, is said to reach more than 1.5 centimeters in length when fully expanded. Since the average flea runs about two to four millimeters in length, you can readily realize that this is quite a remarkable size for a flea. In the East [U.S.A.], we have only one species, *Chaetopsylla lotoris*, which is commonly found on wild carnivores, particularly the raccoon and the gray fox. While not approaching the size of *V. alakurt*, it is quite a large flea.

Bat fleas form a distinct family, the Ischnopsyllidae. In the Northeast [U.S.A.], we have only three bat fleas. The life history of *Myodopsylla borealis*, which was described only a few years ago, largely unknown, but the others represent an interesting contrast in strategy.

Our Eastern [North American] bats, with few exceptions, winter in caves or attics or other protected sites, and breed in summer colonies which, nowadays, are usually in buildings - attics of large buildings - lofts of old barns, and such places. *Myodopsylla insignis* is a parasite of the little brown bat, *Myotis lucifugus*, and less commonly other *Myotis*. In this species, [flea] reproduction occurs in the [bat's] breeding colony. Two recent studies, one by Mary Ellen Tracy at Boston University and the other by Stephen Smith at Ohio State University, have given us a fairly good knowledge of this species, though their accounts differ in one way which will become evident in a moment. Eggs laid at the time of breeding fall from the host into the guano under the bats hanging from the rafters. Here the flea develops through the larval stage, requiring both blood and bat guano to mature. The blood presumably comes from adult flea above, squirting out feces which consist of partially digested blood. After pupation, the adult flea emerges, and is then presented with a major problem. How does it get to the bats, several feet away vertically? One way would be for them to crawl away from the bats to the eaves, then up a slanting rafter till they reach a bat. Some authorities believe this is the method of choice. In a recent paper, it has been demonstrated that a closely related European species moves upward in a different manner. In the breeding colonies, young bats often fall from the rafters into the guano. Adult bats, presumably the parents, rescue such fallen bats within a short time, but in the meantime, several fleas have found them and are thus carried up to the rafters where they can find abundant food. If the bat is not rescued, the fleas will at least be able to feed on it as long as it remains alive, which will give them enough nourishment so that they can breed. This process has not been observed in our *Myotis*, but it seems logical that it does occur. [9]

Now comes the unanswered question on which Tracy and Smith disagree. What happens between the end of one breeding season and the onset of the next one? Tracy states that adults accompany the bats to their wintering quarters and return with them in the spring to start a new generation. Smith thinks that they probably survive as pupae, overwintering in the guano and coming out when bats return in the spring. He bases this opinion on 2 facts: 1) He found very few fleas on

hibernating bats, and 2) a large number of fleas appeared almost at once when bats returned to their breeding sites. I suspect that both are wrong, and that some adults overwinter in crevices in the rafters and emerge in spring to feed and lay eggs. In my experience, few if any fleas are found on bats in hibernating quarters. I and my students have collected hundreds, but have found significant numbers of fleas only in the breeding sites. It would seem possible to remove bats from a cave and take them to a known summer breeding site, and see if they pick up fleas there and if so how fast. No one to my knowledge has done this.

The second Eastern [North American] flea about which we know a good deal is *Nycteridopsylla chapini*, a parasite of the big brown bat, *Eptesicus fuscus*. This flea spends the entire life in the wintering caves. Big brown bats in breeding colonies are flealess, at least in my rather extensive experience. Adult fleas live in crevices in the cave walls and emerge to feed periodically. Eggs are laid in these crevices, or sometimes when the flea is on a host. If the eggs fall to the floor they are probably lost. In the crevices, there is abundant blood in the form of flea droppings, and these larvae do not require bat guano to develop. When these fleas emerge as adults, if they do so during the hibernating season, they will be within inches of a suitable host. Again, we do not know how they over-summer, but it is likely to be as either pupa or adult. We owe our knowledge of this flea largely to Nixon Wilson, of Northern Iowa University, who studied them in caves in Indiana. A Czech scientist, Hurka, has studied a species with the same life history pattern as this in Europe, and he was unable to find any evidence of the fleas in any stage in summer in the caves. Of course, if pupae were deep in crevices, they would be virtually impossible to find, whereas adults would be more likely to be seen, which might suggest that the pupa is the more probable over-summering stage. [10]

I have left to the last the most complex situation yet studied, and the one which I have the most direct knowledge, since my students and I have been working on it for nearly ten years. It involves four species of fleas, all of which infest the southern flying squirrel, *Glaucomys volans*, in the Northeast [U.S.A.]. Some ten years ago one of my students trapped a flying squirrel in a forest near the college where I teach. Fascinated by these pretty little creatures, he read up on them and found that they will readily use nest boxes. So he built a half dozen nest boxes and put them on trees in the area where the squirrel had been caught. To his delight, he soon had squirrels using his boxes. His preliminary work demonstrated the presence of four species of fleas, and later on a student named Jon Day studied their life histories and erected a couple of dozen boxes [11]. After Jon completed his study, Michael Surman attempted laboratory rearing studies on these fleas [11]. Since then I have tried to fill in some of the gaps which remain in our data. Unfortunately there are still some gaps left.

It is quite easy to begin with collection of eggs, because eggs of each species can be readily recognized. *Opisodasys pseudarctomys* is a fairly large flea - this is a male - and lays large oval eggs with little surficial [sic, superficial] character. This egg is not in perfect shape, but may give you some idea. *Epitedia faceta* is a small flea, and lays smaller eggs which are somewhat urn-shaped. *Conorhinopsylla stanfordi* is a rather odd-looking flea, especially the male. Its name means cone-nosed flea, which is not entirely inappropriate. Its eggs are beautifully sculptured in a reticulated pattern which makes them easily recognizable. The fourth flea is *Orchopeas howardi*, a parasite of all kinds of squirrels - gray, red, fox, chipmunk - as well as the flying squirrel. I have no pictures of its eggs, but they are small, oval and very smooth.

In the laboratory, under 20 degrees Celsius conditions, eggs of three of these species hatch in less than a week. Larval periods vary but are about a week or so under these conditions. Pupal periods are relatively uniform at about seven or eight days. Thus an average cycle from oviposition to emergence of the imago would be about three weeks, give or take a few days.

In nature the situation is not quite so simple. If we collect an occupied nest in January, adults will be largely *Orchopeas howardi* with small numbers of each of the other species. If, however, we keep the nest in the laboratory large numbers of *Opisodasys pseudarctomys* will emerge in about ten days, indicating that they were in the pupal stage when the nest was brought in. In nature, this emergence occurs very suddenly and in great numbers about the end of February or early March, after a few warm days. If we collect a nest in February, practically all adults are *Orchopeas howardi*, but again large numbers of *Opisodasys* will emerge in a few days. If we sift such nest material carefully and extract larvae and eggs, most of the larvae will be *Conorhinopsylla* and the only viable eggs will be those of *Epitedia faceta*. [12]

I construe this to mean that *Orchopeas howardi* overwinter in the adult form, and reproduce near the end of winter to produce an immense new generation in late March or April. *Opisodasys pseudarctomys* overwinter in the pupal form, with a large emergence taking place in very early spring. *Epitedia faceta* overwinter in the egg, while the larvae of *Conorhinopsylla stanfordi* overwinter. *Orchopeas* has at least four or five generations per year, *Opisodasys* has three or four. I have taken good numbers of adults in late summer in Minnesota and northern New York, but not around western New York, so I am not sure if a fourth generation per year occurs here in the summer. [13]

We feel that we have the annual cycles of these two species pretty well in hand. The other two present a problem. *Conorhinopsylla* appears to have but one adult generation per year emerging through the fall. I find this rather hard to believe, and it is quite possible that they work in two generations in this fall and winter period. However, from then on adults are almost never taken. They overwinter as larvae, and I have maintained larvae for up to sixty days in an outdoor box in my laboratory window. I was unable to rear any of these larvae to adulthood, however, so I do not know what happens to them from then to late September. One might suspect that they pupate in spring, and spend the summer as pupae. But I simply don't know.

Epitedia faceta overwinter, as I have said, as eggs. I have been able to collect viable eggs of this species as late as March, when they were the only viable eggs I could find in the nests. Of course, a few adults live all or most of the winter, but I think most of the eggs are laid in early winter. If these eggs produce a spring generation, we have not been able to detect it. [14]

Just when I left Fredonia, I collected a nest which had many adult fleas. I did not have time to look at very many, but among the 15 or so that I collected there was at least 1 *Epitedia*. Perhaps I will find a spring generation of adults in this nest when I return.

Our model is not without flaws, as we fully realize. When we extract nest material from a box, removing all parasites in the process, we alter the future events in that box. When we try to rationalize data from field collections, data from laboratory studies and data from nests maintained under more or less natural outdoor conditions but in the laboratory, we run into difficulties. However, when data

secured in all these ways point to the same general conclusions, we think that these conclusions have some validity as far as they go.

Just a week or two ago, I read that the incidence of bubonic plague was increasing in the western United States. Since fleas are the vectors of this often fatal disease, its increase emphasizes the importance of understanding flea life histories. I hope that what I have said today may spark an interest in the study of fleas in at least one or two of you, and you will be able to have as much fun and satisfaction from that study as I have had over the past 35 years.

Thank you.

- A.H. Benton

Annotations of R.L. Bossard to A.H. Benton's "Life History Strategies Among the Siphonaptera"

1. Flea larvae species show a variety of diets, feeding behaviors, and mandible shapes of complex three-dimensional forms, including chewing with mandibles or sucking fluids with pharynx. Flea larvae can be detritivores on organic matter, carnivores on mites, ticks, etc., or cannibals on flea eggs, other larvae, naked pupae, and adult fleas. *Uropsylla tasmanica* (Lycopsyllidae) larvae feed on their vertebrate hosts (Marshall 1981).

The dietary requirements of flea larvae for blood of specific hosts, fungi (yeast), organic matter, etc. appear rather involved in the literature and dependent on the species and their habitat. *Ctenocephalides* larvae do not eat organic matter (Krämer et al. 2001), but eat blood or adult flea feces, as do *Echidnophaga* larvae; *Xenopsylla* eat organic matter but are helped by yeast vitamins (Sharif). *Tunga monositus* larvae do not feed (Elbel 1991).

Hinkle et al. (1991) hypothesized that "fleas may have initially been scavengers... they have subsequently evolved nearly complete hematophagy" for both adults and parental care of their larva. Silverman and Appel (1994) hypothesized: "host blood excretion [by adult fleas] evolved in response to larval nutritional requirements.", however, Almeida et al. (2020) showed that blood in tick feces was also usable by flea larvae.

A variety of flea larval mandibles illustrated by Elbel and Pilgrim ranged from broad and generalized insect cutting/chewing mandible (*Rhadinopsylla*) to specialized, narrow mandibles, some with 3 or 5 spatulate teeth (*Hystrihopsylla*, Ceratophyllidae). Flea larvae of different species in the same nest may have different mandible shapes. Pilgrim (1988) hypothesized that this had to do with differences in diet. Specialization on food could minimize interspecific competition among flea species (see note 14 below). Also, *Notiopsylla* (Pygiopsyllidae) mandibles change shape as larvae age. Pilgrim (1998) again hypothesized differences in diet. This could minimize intraspecific competition between larvae of different ages. In this regard, there are 2 forms of cat flea feces that Hinkle (1991) hypothesized fed different instars.

Almeida, G. P. S. D., Campos, D. R., Avelar, B. R. D., Silva, T. X. D. A. D., Lambert, M. M., Alves, M. S. R., & Correia, T. R. (2020). Development of *Ctenocephalides felis felis* (Siphonaptera:) in different substrates for maintenance under laboratory conditions. *Revista Brasileira de Parasitologia Veterinária*,

29, e022819.

Elbel, R.E. Siphonaptera. In *Immature Insects*; Stehr, F.W., Ed.; Kendall/Hunt Publishing: Dubuque, IA, USA, 1991; Volume 2, pp. 674–689.

Hinkle, N. C., Koehler, P. G., Kern Jr, W. H., & Patterson, R. S. (1991). Hematophagous strategies of the cat flea (Siphonaptera: Pulicidae). *Florida Entomologist*, 377-385.

Hsu, M. H., Hsu, Y. C., & Wu, W. J. (2002). Consumption of flea faeces and eggs by larvae of the cat flea, *Ctenocephalides felis*. *Medical and veterinary Entomology*, 16(4), 445-447.

Krämer, F., Mencke, N., Krämer, F., & Mencke, N. (2001). Developmental cycle of fleas. *Flea Biology and Control: The Biology of the Cat Flea Control and Prevention with Imidacloprid in Small Animals*, 17-34.

Lawrence, W., & Foil, L. D. (2002). The effects of diet upon pupal development and cocoon formation by the cat flea (Siphonaptera: Pulicidae). *Journal of Vector Ecology*, 27, 39-43.

Marshall, A.G. *The Ecology of Ectoparasitic Insects*; Academic Press: London, UK, 1981.

MOLYNEUX, D. Feeding Behaviour of the Larval Rat Flea *Nosopsyllus fasciatus* Bosc. *Nature* 215, 779 (1967). <https://doi.org/10.1038/215779a0>.

Pilgrim, R. L. (1998). Larvae of the genus *Notiopsylla* (Siphonaptera: Pygiopsyllidae) with a key to their identification. *Journal of medical entomology*, 35(4), 362-376.

Rust, M. K., & Dryden, M. W. (1997). The biology, ecology, and management of the cat flea. *Annual review of entomology*, 42(1), 451-473.

Silverman, J., & Appel, A. G. (1994). Adult cat flea (Siphonaptera: Pulicidae) excretion of host blood proteins in relation to larval nutrition. *Journal of Medical Entomology*, 31(2), 265-271.

2. Larvae of *Uropsylla tasmanica* (Marshall 1981) and some Tungini *Tunga monositus* and *T. penetrans* (Elbel 1991) have only 2 instars.

3. Naked pupae result when cocoon spinning fails.

Dryden, M. W., & Smith, V. (1994). Cat flea (Siphonaptera: Pulicidae) cocoon formation and development of naked flea pupae. *Journal of medical entomology*, 31(2), 272-277.

4. Cat fleas mate and oviposit on the host.

Hsu, M. H., & Wu, W. J. (2001). Off-host observations of mating and postmating behaviors in the cat flea (Siphonaptera: Pulicidae). *Journal of medical entomology*, 38(3), 352-360.

5. Non-sticky eggs fall into the environment: Those of the cat flea tend to fall in host's bedding because the flea's oviposition is synchronized with the host's circadian activity, but the sheep flea *Vermipsylla alakurt* [Family Vermipsyllidae] oviposits black eggs into pastures (Marshall 1981). Sticky eggs may be oviposited off the host where the flea glues them to detritus, *Uropsylla tasmanica* glues its eggs to

the host's fur.

6. Larvae and adults of *Hoplopyllus glacialis glacialis* Tasch. [= *Euhoplopyllus glacialis glacialis* (Tasch. 1880)] [Family Pulicidae] were in the fur of all the arctic hare *Lepus arcticus groenlandicus* Rhoads from East Greenland. It's unclear to me how the larvae stay in the fur or what they eat.

Similarly, for *Glaciopsyllus antarcticus* (Ceratophyllidae), "Larvae were blood feeders and pupated amongst the down of host chicks", whilst adult fleas likely overwintered on the bird as it flew over the ocean!

Bell, P.J., Burton, H.R. & van Franeker, J.A. Aspects of the biology of *Glaciopsyllus antarcticus* (Siphonaptera: Ceratophyllidae) during the breeding season of a host (*Fulmarus glacialis*). Polar Biol 8, 403–410 (1988). <https://doi.org/10.1007/BF00264716>

FREEMAN, R., MADSEN, H. A Parasitic Flea Larva. Nature 164, 187–188 (1949). <https://doi.org/10.1038/164187b0>

7. Both "larvae and adult male of *T. monositus* do not feed" (Linardi et al. 2014). So the intake of biomass into the flea population rests entirely on the burrowing female adult who lays large eggs (Elbel 1991)!

Linardi PM, Beaucournu J-C, de Avelar DM & Belaz S: 2014. Notes on the genus *Tunga* (Siphonaptera: Tungidae) II – neosomes, morphology, classification, and other taxonomic notes. Parasite, 21, 68.

8. "Hypertrophied" female fleas may be under the host's skin and sessile (e.g., "*Tunga* (Tungidae) and *Neotunga* (Pulicidae)", permanently attached upon the host's skin and sessile ("*Echidnophaga* (Pulicidae), *Hectopsylla* ((Tungidae))", or temporarily attached upon the host's skin and non-sessile, i.e., "*Chaetopsylla* pro parte, *Dorcadia*, *Vermipsylla* (Vermipsyllidae), *Parapsyllus* pro parte (Rhopalopsyllidae), *Glaciopsyllus* (Ceratophyllidae), *Ancistropsylla* (Ancistropsyllidae), *Malacopsylla* and *Phthiropsylla* (Malacopsyllidae)" (Linardi et al. 2014). It is not clear to me that all of these genera are showing "neosomy" which requires cuticular growth, and not merely expansion.

Rothschild (1992) commented that "the sessile life-style is an evolutionary 'after-thought'"., however, Whiting et al. (2008) hypothesized that "neosomy is actually a primitive rather than a derived lifestyle within fleas".

Hypertrophy might be advantageous when the food scavenged by larvae is poor or unreliable.

Rothschild, M. (1992). Neosomy in fleas, and the sessile lifestyle. Journal of Zoology, 226(4), 613-629.

Whiting, M. F., Whiting, A. S., Hastriter, M. W., & Dittmar, K. (2008). A molecular phylogeny of fleas (Insecta: Siphonaptera): origins and host associations. Cladistics, 24(5), 677-707.

9. There seem to be at least 5 ways that bat fleas in guano on the floors, walls, and ceilings of the roost could reach bats hanging from the ceiling of the roost.

i. Bat fleas such as *Sternopsylla distincta* are slow walkers and weak jumpers, but have long

limbs and can climb up a steep incline in the laboratory (negative geotropism). This climbing ability may be sufficient to bring them to the ceiling (Elbel & Bossard 2007), however Hutson (1981) noted, in regard to bat flea *Ischnopsyllus simplex*, "There was no evidence of any of these fleas making any attempt to climb". In *Myodopsylla insignis*, jumping depended on whether or not the flea had reached the host (Smith and Clay 1988). Climbing and jumping in bat fleas may be plastic behaviors: Unfed fleas climb and jump, fed fleas do not. Or these behaviors may differ in different species.

Unknown stimuli of unknown organs of bat fleas likely triggers pupal emergence and host-locating behaviors (Smith and Clay 1988, Elbel & Bossard 2007).

ii. Bats occasionally brush up or even collide against barriers as they fly, and there are "groundings" by young bats learning to fly (Hutson 1981, Orbach and Fenton 2010, Sändig et al. 2014). When bat fleas *Sternopsylla distincta* ascend to the highest point they can (such as onto the tip of a stalagmite), they quest (Elbel et al. 1999, Elbel & Bossard 2007). If bats brush by, it seems likely the flea could latch on.

iii. As Benton describes, bats often retrieve fallen pups (Kunz & Hood 2000, Bohn et al. 2009), presumably along with fleas from the floor of the roost.

iv. Earwigs (Dermaptera), showing negative geotropism (Hutson 1981), carry bat fleas upward in an unusual phoresy in Indo-Malaysian caves (Hastriter et al. 2017). An earwig carrying fleas possibly could also grapple onto bats.

Phoresy is important for fleas in order to balance the conflicting demands on a metapopulation between dispersal and host-specificity (Bartlow & Agosta 2021, cf. bat fly: Dick & Patterson 2007, beetle: Ohnishi et al. 2021). "Accidental" vertebrate hosts are frequently found with fleas; even if the flea is unable to complete its life cycle on them, the flea could be transported to new and improved environments, e.g., carnivores transporting rabbit fleas (Iannino et al. 2017), contra Belthoff et al. 2021. Fleas themselves carry phoretic mites (Houck & OConnor 1991, Hastriter & Bush 2014, Bezerra-Santos et al. 2021).

v. Finally, some eggs may be laid in ceiling crevices so the fleas are already near to the bats. All of these behaviors could use further observations.

Bartlow, A. W., & Agosta, S. J. (2021). Phoresy in animals: review and synthesis of a common but understudied mode of dispersal. *Biological Reviews*, 96(1), 223-246.

Belthoff, J. R., Elgin, A. A., Navock, K. A., & Bernhardt, S. A. (2021). Burrowing owls as potential phoretic hosts of ground squirrel fleas during a plague epizootic. *Journal of Vector Ecology*, 46(1), 48-56.

Bezerra-Santos, M. A., Sgroi, G., Mendoza-Roldan, J. A., Khedri, J., Camarda, A., Iatta, R., ... & Otranto, D. (2021). Ectoparasites of hedgehogs: From flea mite phoresy to their role as vectors of pathogens. *International Journal for Parasitology: Parasites and Wildlife*, 15, 95-104.

Bohn, K. M., Moss, C. F., & Wilkinson, G. S. (2009). Pup guarding by greater spear-nosed bats. *Behavioral Ecology and Sociobiology*, 63, 1693-1703.

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- Elbel, R.E., R.L. Bossard, M.W. Dryden, J.M. Vincent, & P.M. Bradley. 1999. Observations on the bat fleas, *Myodopsylla insignis* (Rothschild) and *Sternopsylla distincta* (Rothschild). *Journal of the Utah Academy of Science, Arts, and Letters Encyclia* 76: 140-144.
- Elbel, R.E., & R.L. Bossard. 2007. Observations and larval descriptions of fleas (Siphonaptera: Ceratophyllidae, Ctenophthalmidae, Ishnopsyllidae) of the southern flying squirrel, little brown bat, and Brazilian free-tailed bat (Mammalia: Rodentia, Chiroptera). *Journal of Medical Entomology* 44: 915-922.
- Hastriter, M. W., & Bush, S. E. (2014). Description of *Medwayella independencia* (Siphonaptera, Stivaliidae), a new species of flea from Mindanao Island, the Philippines and their phoretic mites, and miscellaneous flea records from the Malay Archipelago. *ZooKeys*, (408), 107.
- Hastriter, M.W., Miller, K.B., Svenson, G.J., Martin G.J., & Whiting M.F. (2017) New record of a phonetic flea associated with earwigs (Dermaptera, Arixeniidae) and a redescription of the bat flea *Lagaropsylla signata* (Siphonaptera, Ischnopsyllidae). *ZooKeys* 657:67–79.
- Houck, M. A., & OConnor, B. M. (1991). Ecological and evolutionary significance of phoresy in the Astigmata. *Annual review of entomology*, 36(1), 611-636.
- Hutson, A. M. (1981). Observations on host-finding by bat-fleas, with particular reference to *Ischnopsyllus simplex* (Siphonaptera; Ischnopsyllidae) in Great Britain. *Journal of Zoology*, 195(4), 546-549.
- Iannino, F., Sulli, N., Maitino, A., Pascucci, I., Pampiglione, G., & Salucci, S. (2017). Fleas of dog and cat: species, biology and flea-borne diseases. *Veterinaria Italiana*, 53(4), 277-288.
- Kunz, T. H., & Hood, W. R. (2000). Parental care and postnatal growth in the Chiroptera. *Reproductive biology of bats*, 415-468.
- Ohnishi, O., Takenaka, M., Okano, R., Yoshitomi, H., & Tojo, K. (2021). Wide-scale gene flow, even in insects that have lost their flight ability: Presence of dispersion due to a unique parasitic ecological strategy of piggybacking hosts. *Zoological Science*, 38(2), 122-139. doi:10.2108/zs200088 [*Meloe* beetle]
- Orbach, D. N., & Fenton, B. (2010). Vision impairs the abilities of bats to avoid colliding with stationary obstacles. *PLoS One*, 5(11), e13912.
- Sändig, S., Schnitzler, H. U., & Denzinger, A. (2014). Echolocation behaviour of the big brown bat (*Eptesicus fuscus*) in an obstacle avoidance task of increasing difficulty. *Journal of Experimental Biology*, 217(16), 2876-2884.
10. Each of the 4 genera of bat fleas in temperate North America (north of Mexico) - *Hormopsylla*, *Myodopsylla*, *Nycteridopsylla*, and *Sternopsylla* - (Lewis and Lewis 1994) occupy a unique niche

which may be determined by whether the host is a migrator (Molassidae) or a hibernator (Vespertilionidae), and the thermal niche (cool or warm) (Bossard 2017). For example, *Sternopsylla distincta* overwinters as pupae in temperate caves abandoned by migrating Molassidae (Elbel and Bossard 2007), but some fleas might also survive as adults on bats which have migrated to winter roosts in the south (Bossard 2017).

Several have commented on bat flea rarity. Lewis (1964) noted: "distribution of bat fleas is one of the most enigmatic problems in the entire field... Doubtless the rarity of bat fleas in this area is in part attributable to lack of collecting. Even in areas that have been thoroughly investigated, however, these fleas are still uncommon",

Marshall (1981) remarked: Only 4% of bat species in Panama, and 6% in Malaysia have fleas, so "conditions in the host's home are presumably critical" [it is unclear if temperate bats are more infested].

Chilton (et al. 2000): "The life cycles of most of the 131 named taxa of bat fleas (Siphonaptera: Ischnopsyllidae) are unknown, but they seem to have particular affinity for hosts that are gregarious and that occupy permanent roosts, such as buildings or caves (Lewis and Lewis, 1994). These sites permit the accumulation of bat guano beneath the roost, which is essential for larval development and completion of the life cycle (Lewis and Lewis, 1994)", "Frequency of infestation may be related to the host's roosting behaviors. All species of ectoparasitic insects found in this study spend some portion of their life cycle off the host (Marshall, 1982). Frequent roost-switching and small colony size may result in low infestation rates (Lewis, 1996)", and "many *M. lucifugus* roost in buildings or in deep cavities in trees (Fenton and Barclay, 1980; Crampton and Barclay, 1998), favoring the accumulation of guano necessary for completion of the flea's life cycle. However, the feces of bats that roost beneath loose bark will generally fall far from the roost, and thus these bats will likely experience lower infestation prevalences of fleas." (Chilton et al. 2000).

Different roosts have different advantages for the bat and flea. In addition to physical challenges, I wonder if some roosts could have predators or pathogens on fleas such as mites, pseudoscorpions, or fungi especially in tropical areas, or lack the unusual phoresy when earwigs carry fleas.

Glen Chilton, Maarten J. Vonhof, B. V. Peterson, and Nixon Wilson. 2000. Ectoparasitic Insects of Bats in British Columbia, Canada. *J. Parasitol.*, 86(1), p. 191–192.

Lewis, R.E. (1964). Additional records of fleas from Lebanon, Syria and Jordan. *Proceedings of the Royal Entomological Society London Series A, General Entomology* 39: 157–165.

Lewis, R. E., & Lewis, J. H. (1994). Siphonaptera of North America north of Mexico: Ischnopsyllidae. *Journal of medical entomology*, 31(3), 348-368.

Bossard, R.L. 2017. Population model for the bat flea *Sternopsylla distincta* (Siphonaptera: Ischnopsyllidae) in temperate caves of the Brazilian free-tailed bat *Tadarida brasiliensis* (Mammalia: Chiroptera). *Journal of the Utah Academy of Sciences, Arts, & Letters* 94: 71-78.

11. Benton, A.H., and J.F. Day. 1980. Seasonal changes in the flea fauna of nests of the southern flying squirrel. pp. 401-403. In: Proceedings of the International Conference on Fleas, Edited by R. Traub and H. Starcke. Ashton Wold, Peterborough, UK, 21-25 June, 1977.
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12. One of the advantages of working with the flying squirrel flea community is that eggs, larvae (Elbel and Bossard 2007), and adults are now identifiable to species. Pupae may be identified by letting them emerge as adults. Benton did not count immature stages to my knowledge, but these data would enlighten the life histories involved.
13. Benton hypothesizes that each species in a flea community overwinters in a different stage. It is untested to what extent this describes other flea communities, but nests of diverse rodent species often have 4 or 5 dominant flea species, seasonally spaced (Bossard 2022).

We might predict that *Orchopeas howardi* adults would occur as broadly temperature tolerant including the coldest time of the year, *Opisodasys pseudarctomys* in hottest (summer) temperatures, *Epitedia faceta* and *Conorhinopsylla stanfordi* intermediate (spring and fall) temperatures. This tests the hypothesis that each species is adapted to a different thermal niche (Bossard 2022). Indeed, *Orchopeas howardi* occurred at coldest or hottest temperatures of the year, but with an absence at intermediate temperatures (Bossard 2022). However, *Opisodasys pseudarctomys* did not occur much during hottest temperatures, falsifying this prediction. Interestingly, *Opisodasys keeni* in deer mice nests was at the hottest. *Epitedia faceta* and *Conorhinopsylla stanfordi* were intermediate, thus corroborating prediction.

Bossard (2022) also hypothesized that *Orchopeas howardi* and *Opisodasys pseudarctomys* represented an evolutionarily newer, more abundant Ceratophyllidae community of a host generalist and a host specialist that was superimposed over an older, less abundant Ctenophthalmidae community of a host generalist and a host specialist (*Epitedia faceta* and *Conorhinopsylla stanfordi*). Although the

players change, the play remains the same?

14. Both Benton and Bossard hypothesized that competition between flea species in nests was significant, and that the seemingly different life histories and thermal preferences were adaptations that minimized competition. However, this hypothesis has not been rigorously tested. There are multiple alternative hypotheses on flea community assembly including dispersal, ecological fitting, environmental filtering, neutrality, and phylogenetic constraints (Bossard 2022, Krasnov et al. 2022).

Bossard, R.L. 2022. Thermal niche partitioning and phenology of Nearctic and Palearctic flea (Siphonaptera) communities on rodents (Mammalia: Rodentia) from five ecoregions. *Journal of Vector Ecology* 47 (2): 217-226.

Krasnov, B.R., Shenbrot, G.I., Khokhlova, I.S. (2022) Regional flea and host assemblages form biogeographic, but not ecological, clusters: evidence for a dispersal-based mechanism as a driver of species composition. *Parasitology*, 149 (11), pp. 1450-1459.

R.L. Bossard

Siphonaptera Literature (2024)

New Flea Species

(2024)

Acosta, R., Morrone, J. J., Hernandez, C. F., & Fernandez, J. A. (2024). List of the flea species of the State of Chihuahua, Mexico, with new records and a new species of *Strepsylla* Traub, 1950 (Siphonaptera: Ctenophthalmidae: Neopsyllinae: Phalacropsyllini). *Zootaxa*, 5410(2), 239-253.

A total of 45 flea species, assigned to 22 genera, are documented for the state of Chihuahua, Mexico. Fourteen records are new for the state, five species are recorded for the first time for the country and the state, and the number of species of Mexico is increased to 177 species. Chihuahua now has 25.4% of the total flea species registered for the country. A new species of the genus is described from specimens from the Flora and Fauna Protection Area Cerro del Mohinora, in the northwestern part of the Sierra Madre Occidental. *Strepsylla mohinora* sp. nov. is included in a published taxonomic key for the species of *Strepsylla*

[*Strepsylla mohinora* sp. nov. Acosta et al. 2024]

Hastriter, M. W. (2024). Revision of the Flea Genus *Ctenidiosomus* (Siphonaptera: Pygiopsyllidae) with Descriptions of Four New Species from Costa Rica and Peru. *Annals of Carnegie Museum*, 90(2), 119-138.

Four new species of fleas of the genus *Ctenidiosomus* Jordan, 1931, are described from Puntarenas and San Jose provinces, Costa Rica and the Piura and Cuzco regions of Peru (*Ctenidiosomus eckerlini*, *Ctenidiosomus churchi*, *Ctenidiosomus pattersoni*, and *Ctenidiosomus solusensus*). This is the first record of the genus in Central America. Four specimens of *C. eckerlini* were collected from three different host species (*Peromyscus mexicanus* (Saussure, 1860), *Nephelomys*

devius (Bangs, 1902) and one from a composite of three different hosts (none of which it had ever been collected from previously). The preferred host for the new Peruvian species *C. churchi*, appears to be *Thomasomys cinereus* Thomas, 1882, although additional collections are needed to validate its true host. *Ctenidiosomus pattersoni* occurred on *Lestoros inca* (Thomas, 1917) but was most abundant on *Akodon torques* (Thomas, 1917). The fourth new taxon, *C. solusensus*, was described from two females, one collected from *Akodon aerosus* Thomas, 1913, and the other from *Nephelomys levipes* (Thomas, 1902). All known records of *Ctenidiosomus* were from species belonging to the rodent family Cricetidae with two exceptions from an opossum shrew, *Caenolestes fuliginosus* (Tomes, 1863) and *L. inca*. Table 1 identifies the documented host species for the nine known species. The genus *Ctenidiosomus* was known previously only from South America (Venezuela to Argentina) and is the only member of the family Pygiopsyllidae in the Western Hemisphere. Except for *Ctenidiosomus perplexus* Tipton and Machado-Allison, 1972, there are only a few records for each of the other eight species of this geographically enigmatic genus that spans the entire length of the western Neotropical Region. Considering the diversity of host records (n = 23), individual species have seldom been recovered from the same host species. The presence of setae on the aedeagus is pointed out for the first time in the genus. A discussion of the origin, phylogenetic relationship to Australian pygiopsyllids, its presence in the New World, and host associations for the genus are discussed; however, conclusive evidence remains unclear and requires further investigation.

[(*Ctenidiosomus eckerlini* Hastriter 2024, *Ctenidiosomus churchi* Hastriter 2024, *Ctenidiosomus pattersoni* Hastriter 2024, and *Ctenidiosomus solusensus* Hastriter 2024]

Keskin, A. (2024). *Nosopsyllus (Nosopsyllus) spiniformis* n. sp., a new ceratophyllid flea (Siphonaptera, Ceratophyllidae, Ceratophyllinae) from Trkiye. Zootaxa, 5424(3), 389-395.

[Abstract in FN 93]

[*Nosopsyllus (Nosopsyllus) spiniformis* n. sp. Keskin 2024]

Li, F., Zhao, S., Li, E., Xie, S., Wang, N., Tan, W., & Wang, Y. (2024). Detection of *Rickettsia raoultii* in *Vermipsylla alakurt*-Like Fleas of Sheep in Northwestern China. Acta Parasitologica, 69(1), 776-784.

Introduction: To date, a total of 2574 validated flea species have been discovered. Vermipsyllidae is a family of fleas that comprises at least eight species. *Vermipsylla* is a genus of the family Vermipsyllidae within the order Siphonaptera of fleas. Here a novel *Vermipsylla* species was described, and rickettsial agent was also detected in it.

Methods: A total of 128 fleas were collected directly from 260 pastured sheep in China. Of these, eight representative fleas (four males and four females) were identified by key morphological features. Meanwhile, 120 flea DNAs, including six flea samples for molecular taxonomy, were subjected to *Rickettsia* spp. DNA detection. The molecular identity of fleas was determined by amplification and sequence analysis of four genetic markers (the 28S rDNA genes, the 18S rDNA genes, the mitochondrial cytochrome c oxidase subunit I and subunit II). In addition, five *Rickettsia*-specific gene fragments were used to identify the species of the rickettsial agents. The amplified products were sequenced and phylogenetically analyzed.

Results: The morphological characteristics of the flea species identified in this study were similar to *Vermipsylla alakurt*, but presented difference in hair number of the metepimeron, the third tergum, the genitals and the tibiae of hind leg. The 18S rDNA, 28S rDNA and COII genetic markers from fleas showed the highest identity to those of *V. alakurt*, shared 98.45% (954/969), 95.81% (892/931) and 85.86% (571/665) similarities, respectively. However, the COI sequence showed the

highest identity to that of *Dorcadia ioffi* with 88.48% (576/651) similarity. *Rickettsia raoutii* tested positive in 14.17% (17/120) flea DNA samples.

Conclusion: Our study reports the detection of *R. raoutii* in *V. alakurt*-like fleas infesting sheep in China.

Taxonomy

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Kwak, M. L., Burns, P. A., Hastriter, M. W., Jenkins Shaw, J., & Nakao, R. (2024). The threatened Pookila (*Pseudomys novaehollandiae*) hosts a diverse macrobiome of arthropods at varying risks of co-extinction. *Journal of Insect Conservation*, 1-10.

The Pookila (*Pseudomys novaehollandiae*) is a threatened Australian mammal emblematic of the global decline of many vertebrate taxa. While conservationists are concerned with the survival of threatened vertebrates, few consider the rich diversity of symbiotic arthropods which often depend on these hosts for survival. Using *P. novaehollandiae* as a case study, we demonstrate that even opportunistic sampling can rapidly expand knowledge of the species richness of arthropod symbiont communities on poorly studied vertebrate hosts. We also demonstrate how DNA barcoding can be integrated to provide rapid and inexpensive identification capabilities for non-specialists. Finally, we show how various metrics and methods can be used to quantify and qualify extinction risk of poorly known arthropod symbionts to provide the foundations for conservation action. Implications for insect conservation: This study highlights how opportunistic sampling coupled with DNA barcoding and extinction risk metrics can be used to rapidly and inexpensively characterise the arthropod macrobiomes of rare vertebrates and evaluate the risk of decline of their component taxa.

Keywords Parasite conservation · Australia · Laelaps · Siphonaptera · Ixodidae · New holland mouse

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Hosts and Distribution

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Begam, R., Devi, P., Bhattacharjee, K., Bora, S., Thakuria, N., & Deka, A. A comparative morphological study of flea recovered from dog and cat in Guwahati. DOI: <https://doi.org/10.33545/26174693.2024.v8.i3Sg.818>

A study was conducted to investigate the prevalence of fleas in dogs and cats in Guwahati city from August 2020 to July 2021. A total of 244 dogs and 120 cats were examined, out of which 55 dogs and 24 cats were found to be infested with fleas. For morphological identification, 388 fleas and 252 fleas were collected from infested dogs and cats respectively. Fleas were highly prevalent in stray dogs (76.0%) and cats (62.4%) than in pet dogs (17.5%) and cats (13.2%). The prevalence of flea infestation was higher in young dogs (51.4%) and cats (36.3%) than in adult dogs (37.1%) and cats (32.3%). Two species of fleas were identified namely *Ctenocephalides canis* and *Ctenocephalides felis*. On dogs, the prevalence of *C. canis* alone was 62.0%, whereas that of *C. felis* alone was 38.0%. Prevalence of *C. felis* only on cats was 90.0% whereas that of *C. canis* alone was 10.0%. Mixed infestation of *C. canis* and *C. felis* were recorded in 10 dogs and 2 cats. These two flea species *C. canis* and *C. felis* infesting the host causes irritation, dermatitis and also act as a vector of many diseases, therefore the proper control of these flea is necessary in dogs and cats to prevent flea borne diseases.

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Co-invasion, characterized by the simultaneous introduction of hosts and parasites with the

latter establishing themselves in native hosts, is a phenomenon of ecological concern. *Rattus rattus*, a notorious invasive species, has driven the extinction and displacement of numerous avian and mammalian species and serves as a key vector for diseases affecting both humans and wildlife. Among the parasites hosted by *R. rattus* are fleas, which exhibit obligate parasitic behaviour, a generalist nature and high prevalence, increasing the likelihood of flea invasion. Simultaneously, invasive species can serve as hosts for native parasites, leading to potential amplification or dilution of parasite populations in the environment. In Chile, *R. rattus* has been present since the 17th century because of the arrival of the Spanish colonizers through the ports and has spread throughout urban, rural and wild Chilean territories. This study aims to evaluate whether co-invasion of native fleas of invasive rats occurs on native rodents in Chile and to determine whether black rats have acquired flea native to Chile during their invasion. For this, we captured 1132 rodents from 26 localities (20° S–53° S). *Rattus rattus* was found coexisting with 11 native rodent species and two species of introduced rodents. Among the native rodents, *Abrothrix olivacea* and *Oligoryzomys longicaudatus* exhibited more extensive sympatry with *R. rattus*. We identified 14 flea species associated with *R. rattus*, of which only three were native to rats: *Xenopsylla cheopis*, *Leptopsylla segnis* and *Nosopsyllus fasciatus*. These three species presented a higher parasite load in black rats compared to native fleas. *Leptopsylla segnis* and *N. fasciatus* were also found associated with native rodent species that cohabit with *R. rattus*. The remaining species associated with *R. rattus* were fleas of native rodents, although they were less abundant compared to those associated with native rodents, except for *Neotyphloceras pardinasi* and *Sphinctopsylla ares*. Although there has been evidence of flea transmission from rats to native species, the prevalence and abundance were relatively low. Therefore, it cannot be definitively concluded that these fleas have established themselves in native rodent populations, and hence, they cannot be classified as invasive fleas. This study underscores *R. rattus*' adaptability to diverse environmental and geographical conditions in Chile, including its capacity to acquire fleas from native rodents. This aspect has critical implications for public health, potentially facilitating the spread of pathogens across various habitats where these rats are found.

[Editor's note: "Invasive species are plants, animals, or pathogens that are non-native (or alien) to the ecosystem under consideration, and whose introduction causes or is likely to cause harm." (<https://www.invasivespeciesinfo.gov/>). In addition, invasive species may have traits such as special reproduction, vagility and dispersal, resource acquisition, stress avoidance or tolerance of toxins, mutualisms, and escape of competitor, predator, parasite, or pathogen.

So non-native *R. rattus* fleas when harmful could be considered invasive, whether or not any transfer onto native rodents. The concept of the invasive species, popularized a century ago by ecologist C. Elton, is controversial as to terminology, threat or value posed by invaders, and management options (Shackleton et al. 2022).
- Editor]

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Musese, L. J., Kitegile, A. S., & Kilawe, C. J. (2024). Ectoparasites of wild rodents in forest sites invaded and uninvaded by *Maesopsis eminii* in Amani nature forest reserve, Tanzania. *International Journal for Parasitology: Parasites and Wildlife*, 24, 100932.

The invasion by *Maesopsis eminii* [tree] reduced the abundance of rodents and parasites.

Parasite infestation was substantial on one rodent species, *Montemys delectorum*.

Echinolaelaps echidninus were common in uninvaded, while *Ctenophthalmus calceatus cabirus* were common in the invaded sites.

Parasites are important component of communities in a forest ecosystem with profound effects on trophic interactions such as food web. Modification of the forest structure (e.g., changes in species composition and abundance of key species) can have a strong impact on the occurrence, diversity, and abundance of parasites, with subsequent repercussions for ecosystem functioning. In this study, we compared the occurrence and abundance of wild rodents' ectoparasites from forest sites invaded and uninvaded by an invasive tree, *Maesopsis eminii* in Amani Nature Forest Reserve, Tanzania. Three large plots (40 m × 100 m) were randomly established in each forest sites invaded and uninvaded by M.

emini. In each plot, 50 Sherman traps were systematically placed at 10 m interval for capturing wild rodents through a capture-mark-recapture technique. Wilcoxon rank sum test was used to compare for differences in the abundance of infested rodents and ectoparasites between the invaded and uninvaded forest sites. A total of 297 individual rodents were captured and screened for ectoparasites, including 174 rodents from uninvaded forest site and 123 rodents from invaded forest site. The number of infested rodents were significantly ($W = 8592$, $P < 0.001$) greater in uninvaded forest site (66.27%) than in the invaded forest site (36.2%). Furthermore, a significant greater number of *Echinolaelaps echidninus* ($W = 1849$, $P < 0.01$) and *Dinopsyllus ellobius* ($W = 2800.5$, $P < 0.05$) ectoparasites were found in uninvaded as compared to the invaded forest sites. The results of this study suggest that the invasion and dominance by, *M. emini* in Amani Nature Reserve has created unfavorable conditions for rodents and ectoparasites and therefore impacting the diversity and function of the forest ecosystem. We recommend prevention of further introduction of the *M. emini* outside their natural range and mitigating the impact of the established *M. emini* in Amani Forest Nature Reserve.

Nielsen, Ó. K., Fuglei, E., Pedersen, Å. Ø., & Skirnisson, K. (2024). Svalbard rock ptarmigan: a first glimpse into parasite infections. *Polar Research*, 43.

Nyema, J., Nath, T. C., Bhuiyan, M. J. U., Islam, S., Islam, K. M., Basir, S., ... & Lee, D. (2024). Morpho-molecular investigation of ectoparasitic infestation of companion animals in Sylhet city, Bangladesh. *Veterinary Parasitology: Regional Studies and Reports*, 47, 100953.

Peter, N., Schantz, A. V., Dörge, D. D., Steinhoff, A., Cunze, S., Skaljic, A., & Klimpel, S. (2024). Evidence of predation pressure on sensitive species by raccoons based on parasitological studies. *International Journal for Parasitology: Parasites and Wildlife*, 24, 100935.

Powers, K. E., Eckerlin, R. P., Sheehy, R. R., & Reynolds, R. J. (2024). A Range-Wide Ectoparasite Survey for Allegheny Woodrats (*Neotoma magister*). *The Journal of Parasitology*, 110(2), 179-185.

Reyes, A. A. (2024). Genes and Fleas (Siphonaptera) of the Long-Tailed Ground Squirrel Across Inner East Asia (Master's thesis, The University of North Carolina at Greensboro). 88 pp.

In host-parasite systems, the interplay of biotic and abiotic factors can alter the abundance and distribution of both host and parasite, profoundly impacting the evolution of both and the consequences for human and wildlife health. Understanding these dynamics holds particular significance in regions like Inner East Asia, where natural foci for plague (a flea borne illness caused by the bacterium, *Yersinia pestis*) continue to persist. However, limited sampling efforts for many mammalian species in this area highlight the need for further research to uncover the factors influencing host-parasite dynamics. The overarching goal of this work is to understand the historical (e.g., glacial cycles) and modern (host, climate, land use) factors that shape the diversity of mammals and their ectoparasites across Inner East Asia. To do this, this project aims to resolve phylogeography and response to past environmental change within the long-tailed ground squirrel *Urocyon undulatus* and to describe how this and a suite of current host and climate factors shape the composition of its associated flea (Siphonaptera) community. Firstly, I examined the phylogeny and historical demography of *U. undulatus* across its range in Mongolia and Russia through a combined analysis of three mitochondrial DNA regions. The objective was to reveal the extent to which late-Pleistocene environmental changes have affected the genetic diversity within *U. undulatus* populations through time. Secondly, I examined the abundance and species richness of fleas on *U. undulatus* across Mongolia through a multi-level

community analysis, with the goal of understanding how host-trait, climate, and land use factors influence flea community composition on host individuals. This work provides new biodiversity knowledge and a baseline for understanding host and parasite dynamics in a system that is especially important diseases such as plague.

Romashova, E. B., Medvedev, S. G., & Kuznetsov, Y. E. (2024). Distribution of the flea *Monopsyllus Sciurorum Sciurorum* (Schrunk, 1803)(Siphonaptera, Ceratophyllidae) in the wild and in fur farms. *Parazitologîa*, 58(4), 338-350.

Rose, R. K. (2024). *Oryzomys texensis* (Rodentia: Cricetidae). *Mammalian Species*, 56(1036), seae001.

Sakalauskas, P., Lipatova, I., Gričiuvienė, L., Ražanskė, I., Snegiriovaitė, J., & Paulauskas, A. (2024). Diversity of Flea Species (Siphonaptera) and Their Vector-Borne Pathogens from Bats (Chiroptera) in Lithuania. *Diversity*, 16(4), 192.

Schott, D., Ribeiro, F. L., Santos, F. N., & Carvalho, R. W. D. (2024). Fleas (Siphonaptera, Latreille, 1825) from Rio Grande do Sul State, Brazil: Species Diversity, Hosts, and One Health Approach. *Vector-Borne and Zoonotic Diseases*, 24(5), 308-320.

Schütte, K., Springer, A., Brandes, F., Reuschel, M., Fehr, M., & Strube, C. (2024). Ectoparasites of European hedgehogs (*Erinaceus europaeus*) in Germany and their health impact. *Parasites & Vectors*, 17(1), 2.

Seidi, S., Tavassoli, M., Malekifard, F., & Ben Said, M. (2024). Molecular Identification, Multigenic Characterization and Phylogenetic Analysis of *Ctenocephalides Canis* Fleas from Humans and Domestic Animals in Iran. *Acta Parasitologica*, 1-12.

Starikov, V. P., Sarapultseva, E. S., Kravchenko, V. N., & Bernikov, K. A. (2024, March). Parasitic arthropods of small mammals in the Surgut sanctuary. In 11th International Scientific & Practical Conference "Culture, Science, Education: Problems and Perspectives" (pp. 442-449).

Thoroughgood, J. T., & Chilton, N. B. (2024). Comparison of flea diversity in the burrows of Richardson's ground squirrels (*Urocitellus richardsonii*) in urban and rural sites in central Saskatchewan, Canada. *Journal of Vector Ecology*, 49(2), R61-R69.

Morphology

Berrizbeitia, M. F. L., & Pérez, M. J. (2024). Evidence of castrations in fleas of small mammals (Chiroptera and Rodentia) from Argentina. *Parasitology Research*, 123(2), 136.

Morphological abnormalities in fleas seem to be common in nature and are under reported in Argentina. In this note, we describe anomalies in two males and one female of *Alectopsylla unisetosa* Mahnert (Ischnopsyllidae) and one of *Polygenis* sp. (Rhopalopsyllidae) male collected from small mammals in the provinces of Neuquen and Salta, Argentina. In all specimens, the anomalies were observed at the level of the genitalia recognized as partial castration. The structures mainly affected were the modified abdominal segments, the aedeagus (in male), and the spermatheca (in female). The

present communication is the first one devoted exclusively to teratogenous fleas in Argentina.

Cayrol, B., Arnoldi, I., Novak, V., Epis, S., Brilli, M., Rahbé, Y., ... & Gabrieli, P. (2024). The development of the piercing mouth during the last molt of the diseases-transmitting aphids and mosquitoes as revealed by synchrotron X-ray microtomography. *Entomologia Generalis*.

Piercing-sucking insects are extremely efficient vectors of animal and plant pathogens. This group is polyphyletic with the piercing organ highly heterogenous in its structure and morphogenesis, adapted to a specific host and feeding mechanism. For instance, hemimetabolous aphids feed on plant sap from nymphs to adults and they renew their stylets at each molt, thanks to specialized secreting glands. Distinctly, holometabolous mosquitoes have two feeding modes: larvae feed in water by filtering and scraping surfaces with their mouths, while female adults can acquire blood from vertebrate hosts. The pupal metamorphosis allows switching from one feeding habit to another. Here, we present a deep characterization of the biogenesis of the adult mouthparts in parthenogenetic females of the pea aphid *Acyrtosiphon pisum* (Harris 1776) (Hemiptera: Aphididae) and in females of the mosquito *Aedes albopictus* (Skuse 1895) (Diptera: Culicidae), investigated using non-invasive X-ray synchrotron-based microtomography. Comparing datasets collected from aphid juvenile and adult stages and from preimaginal and adult stages of the mosquito, we were able to track the morphological changes of secreting glands and the synthesis of the adult stylet in aphid heads and to follow the de novo formation of mosquito mouthparts in pupae. Our study provides a baseline for investigating the evolution and the development of piercing-sucking mouthparts and to better understand how morphogenesis works in insects.

Kawata, A., Ogawa, N., & Yoshizawa, K. (2024). Morphology of the pterothoracic musculature in Paraneoptera and its phylogenetic implication (Insecta: Neoptera). *Journal of Morphology*, 285(6), e21712.

Szczepankiewicz, A., & Simiczyjew, B. (2024). Ovary structure and oogenesis in Heteroptera. *Invertebrate Reproduction & Development*, 1-10.

Zhao, C., Huang, M., Yang, D., & Liu, X. (2024). Comparative Morphology of the Wing Base Structure Illuminates Higher-Level Phylogeny of Holometabola. *Insects*, 15(3), 199.

Physiology

Danchenko, M., & Macaluso, K. R. (2024). Salivary glands of the cat flea, *Ctenocephalides felis*: Dissection and microscopy guide. *Current research in insect science*, 100080.

The cat flea, *Ctenocephalides felis*, is an ectoparasite and a disease vector.

Studies on salivary glands enrich knowledge of vector-pathogen-host interactions.

Pictorial dissection guide facilitates research of flea salivary gland molecules.

We illustrated dynamic changes in salivary gland morphology upon blood feeding.

We complemented existing drawings and histology images of flea internal anatomy.

Fleas are morphologically unique ectoparasites that are hardly mistaken for any other insect. Most flea species that feed on humans and their companion animals, including the cat flea (*Ctenocephalides felis*), have medical and veterinary importance. Besides facilitating blood acquisition, salivary biomolecules can modulate pathogen transmission. Thus, dissection of salivary glands is essential for comprehensive studies on disease vectors like the cat flea. Herein, we present the pictorial dissection protocol assisting future research targeting individual flea organs, for revealing their roles in vector competence and physiology. We provide a comprehensive guide, allowing researchers, even with limited practical experience, to successfully perform microdissection for collecting cat flea salivary glands. Furthermore, the protocol does not require expensive, sophisticated equipment and can be accomplished with routinely available tools. We illustrated expected results with morphological changes of salivary glands upon blood feeding as well as fluorescently stained these organs.

Getahun, M. N. (2024). Livestock-Vector interaction using Volatile Organic Metabolites. *Current Opinion in Insect Science*, 101269.

Hyuga, A., Ouma, P., Matharu, A. K., Krücken, J., Kaneko, S., Goto, K., & Fillinger, U. (2024). Myth or truth: investigation of the jumping ability of *Tunga penetrans* (Siphonaptera: Tungidae). *Journal of Medical Entomology*, 61(1), 261-265.

Female sand fleas (*Tunga penetrans* Linnaeus, 1758, Siphonaptera: Tungidae) cause a severe parasitic skin disease known as tungiasis. *T. penetrans* is a small flea, measuring less than 1 mm in length. The females of this species burrow into the skin of human and animal hosts and mostly affect the feet. This has led to the anecdotal assumption that *T. penetrans*, unlike its relatives in the Siphonaptera family, would have a limited jumping ability potentially not reaching higher body parts. However, there is no data supporting this. This study evaluated the jumping capabilities of *T. penetrans* for height and distance using sticky tapes. The vertical jump of the female *T. penetrans* ranged from 4.5 to 100 mm with a mean of 40 mm whereas the vertical jump of the male *T. penetrans* ranged from 1.2 to 138 mm with a mean of 46 mm. The horizontal jump of the female *T. penetrans* ranged from 18 to 138 mm with a mean of 64 mm and that of the male ranged from 9 to 251 mm with a mean of 80 mm. Based on the literature, fleas of various species have been described as jumping vertically 50–100 times their size and horizontally 5–100 times their size. In this respect, sand fleas appear to have equal expert jumping abilities to their relatives. Their aggregation on people's feet is not likely a result of their poor jumping ability but might be an adaptation to the host's behavior which would require further investigations.

Dhruva K. Karnik, 4-29-2024 Antimicrobial activity of cat flea (*Ctenocephalides felis*) gut proteins on different days after blood feeding Biology (B.S.B.Thesis open access)

Koh, J. S., Baek, S. M., Kim, B., Cho, K. J., & Kim, H. Y. (2024). Comparison of water and terrestrial jumping in natural and robotic insects. *Annals of the New York Academy of Sciences*.

Scanlan, J. L., & Robin, C. (2024). Phylogenomics of the ecdysteroid kinase-like (EcKL) gene family in insects highlights roles in both steroid hormone metabolism and detoxification. *Genome Biology and Evolution*, 16(2), evae019.

Katie A. Weber, 2024. Transcriptional Induction of Immune Pathway Genes and Subsequent Antibacterial Activity in the Digestive Tract of the Cat Flea (*Ctenocephalides felis*). Master of Science in Biology (M.S.) Thesis. Georgia Southern University.

Yang, Z., Ren, J., Lu, S., Feng, Y., Fan, Y., Liu, T. X., & Jing, X. (2024). In vivo functional analysis of the cotton bollworm *Helicoverpa armigera* 24-dehydrocholesterol reductase (HaDHCR24) in phytosterol metabolism. *Insect Science*.

Evolution

Booth, W. (2024). Population genetics as a tool to understand invasion dynamics and insecticide resistance in indoor urban pest insects. *Current Opinion in Insect Science*, 101166.

Many urban pest insects now exhibit a near-global cosmopolitan distribution.

Insecticide resistance is common and geographically widespread.

A paucity of studies address population genetics and insecticide resistance across broad geographic and temporal scales.

Emerging genomic resources offer exciting opportunities to investigate the success of indoor urban pest insects.

Many indoor urban pest insects now show a near-global distribution. The reasons for this may be linked to their cryptic behaviors, which make unintentional transport likely, tied to their reliance on human-mediated dispersal that can result in spread over potentially long-distances. Additionally, numerous species exhibit an array of mechanisms that confer insecticide resistance. Using population genetics, it is possible to elucidate the genetic characteristics that define globally successful indoor urban pest insect species. Furthermore, this approach may be used to determine the frequency and distribution of insecticide resistance. Here, I review the recent literature that utilizes population genetic analyses in an effort to identify the characteristics that help explain the success of indoor urban pests.

Cameron, S. L. (2024). Insect Mitochondrial Genomics: A Decade of Progress. *Annual Review of Entomology*, 70.

de Santis, M. D. (2024). Homoplasy as an Evolutionary Process: An Optimistic View on the Recurrence of Similarity in Evolution. *Biological Theory*, 1-12.

Fu, Y. T., Xun, Y., Peng, Y. Y., Zhang, Y., & Wu, X. (2024). The complete mitochondrial genome of the rodent flea *Nosopsyllus laeviceps*: genome description, comparative analysis, and phylogenetic implications. *Parasites & Vectors*, 17.

Gokhman, V. E., & Kuznetsova, V. G. (2024). Structure and Evolution of Ribosomal Genes of Insect Chromosomes. *Insects*, 15(8), 593.

Excerpt: "The large superorder Antliophora harbors insects that belong to three orders, i.e., Mecoptera, Diptera, and Siphonaptera. However, the number and position of rDNA sites have been examined only in members of the two former taxa. In Mecoptera (scorpionflies or hanging flies), only

the only species *Panorpa vulgaris* Imhoff et Labram (Panorpidae) has been studied. In the male karyotype of *P. vulgaris*, FISH with an 18S rDNA probe yielded four positive signals, i.e., apparent NORs [nucleolus-organizing regions]... "

Mooring, M. S. (2024). Programmed Grooming after 30 Years of Study: A Review of Evidence and Future Prospects. *Animals*, 14(9), 1266.

The programmed grooming model is an evolutionary hypothesis for the regulation of ectoparasite-defense grooming behavior. It proposes that grooming to remove ectoparasites such as ticks is regulated by a type of internal biological clock that has evolved to pre-emptively remove parasites before they can blood feed. Programmed grooming contrasts with the stimulus-driven model, in which grooming is stimulated by direct peripheral irritation from ectoparasites. Since programmed grooming was first proposed in 1992, 26 studies have provided robust support for the model with ungulate hosts and ticks. Additional studies from unaffiliated investigators have evaluated the predictions of the model in different host systems (including rodents and primates) and in a variety of ectoparasites (fleas, lice, and keds). After reviewing the current evidence, I conclude that (1) tests of the programmed grooming model should utilize the established protocol, so that the results can be compared and assessed in light of previous studies; (2) the model predictions should be tailored to the host biology under investigation; and (3) the model predictions should be tailored to the ectoparasite biology, since the efficacy of grooming depends on the parasite. It is hoped that future studies will reveal much more about how grooming helps wild animals to defend themselves against the threat of parasites.

Oliveira, T., da Costa, J. P., Araújo, A. V., Dias, D. E. M., & Dantas, M. A. T. (2024). Mangy armadillos: New records of *Karethraichnus minimum* in extinct quaternary pampatheres of the Brazilian intertropical region. *Journal of South American Earth Sciences*, 144, 105034.

Szpila, K., van de Kamp, T., Sontag, E., Krzemiński, W., Kopeć, K., & Soszyńska, A. (2024). The hidden world of fossil larvae: description and morphological insights of an immature scorpionfly (Mecoptera: Panorpidae) from the Baltic amber. *Zoological Journal of the Linnean Society*, zlae009.

So far, there has been no information of the pre-imaginal stages in the well-preserved fossil record of the Mecoptera. The first and well-preserved mecopteran larva was discovered in Eocene Baltic amber. The application of synchrotron X-ray microtomography enabled the reconstruction of the body structure with high accuracy, providing a comprehensive set of morphological data that classical stereoscopic microscopy could not capture. The larva is eruciform, with distinct segmentation of the body. All the most important morphological structures are documented, including the chaetotaxy system. Shape of antenna and annulated processes on the abdominal segments point to the third/four stage of development. There is no doubt that the larva belongs to Panorpidae, the most abundant family of extant Mecoptera. The morphological characters point to the closest affinity to larvae of *Cerapanorpa* and *Panorpa* but the fossil larva cannot be assigned to any extant genus of Panorpidae. Based on the results of synchrotron X-ray microtomography, it is possible to discuss the assignement of this larva to a lower systematic level than the family, and draw conclusions about the nature of the habitat preferences of Eocene panorpid.

Tian, Y., Gao, H., Li, H., Li, C., & Li, B. (2024). Evolutionary origin and distribution of leucine-rich repeat-containing G protein-coupled receptors in insects. *Comparative Biochemistry and Physiology Part D: Genomics and Proteomics*, 52, 101318.

Xiaoxia, L. I. N., & Wenge, D. O. N. G. (2024). Research advances on Pulicinae mitochondrial genomes. *CHINESE JOURNAL OF PARASITOLOGY AND PARASITIC DISEASES*, 42(3), 389.

Zhang, Y., Rasnitsyn, A. P., Zhang, W., Song, F., Shih, C., Ren, D., ... & Gao, T. (2024). Stem chewing lice on Cretaceous feathers preserved in amber. *Current Biology*, 34(4), 916-922.

Ecology

Стариков, В. П., & Вершинин, Е. А. (2024). БЛОХИ (SIPHONAPTERA) МЕЛКИХ МЛЕКОПИТАЮЩИХ РАЗНОТРАВНО-ДЕРНОВИННО-ЗЛАКОВОЙ СТЕПИ ЮЖНОГО ЗАУРАЛЬЯ (КУРГАНСКАЯ ОБЛАСТЬ). Степи Северной Евразии: материалы X международного симпозиума, (X), 1277-1282. [FLEAS (SIPHONAPTERA) OF SMALL MAMMALS OF FORB-SOD-GRASS STEPPE OF THE SOUTHERN TRANS-URALS (KURGAN REGION)]

The results of the study of fleas of small mammals of the forb-sod-grass steppe of the Southern Trans-Urals for the 1980s-1990s and 2022 in the Kurgan Region are summed up. 20 species and subspecies of fleas have been established. Belonging to different ecological groups. In terms of the number of individuals, the group of fleas dominates, gravitating towards forest, meadow forest complexes and moist biotopes, which is determined by the increased concentration of feeders here in the summer. The work shows the specifics of species diversity, occurrence, abundance and intensity of flea infestation, depending on the method of counting small mammals. According to long-term data, two dominant species of fleas have been established - *Ctenophthalmus assimilis* and *Corrodopsylla birulai*. *Megabothris walkeri*, *Hystrihopsylla talpae* and *Frontopsylla elata* were identified as codominants. The greatest diversity of fleas (9-13 species) fed the background species of small mammals of the forb-sod-grass steppe (common vole, housekeeper vole, red vole, lesser forest mouse and common shrew). For the first time on the territory of the West Siberian Plain, the flea *Ctenophthalmus congeneroides* was recorded, which was found on five (common shrew, steppe pied vole, housekeeper vole, common vole and lesser forest mouse) hosts-feeders. Prospects for further studies of this group of parasitic arthropods are shown.

Daniel, M., & Kalvoda, J. (2024). An Actual Biogeographical Hazard of the Expansion of Small Mammals and Their Parasites Linked with Climate Changes in the East Nepal Himalaya. In *The Nature of Geomorphological Hazards in the Nepal Himalaya* (pp. 325-357). Cham: Springer International Publishing.

Gil-Fernández, M., Vargas-Sandoval, M., Delfín-Alfonso, C. A., Mendoza, E., Godínez-Gómez, O., Jiménez-Lara, N. K., ... & Le Roux, J. J. (2024). Host sweet host: Rodent communities support similar ectoparasite diversity regardless of anthropogenic disturbance. *Journal of Applied Entomology*, 148(5), 537-552.

Horak, Ivan G.; Joop Boomker; Grabovsky, V. I.; Khokhlova, I. S.; Kerstin Junker; Sanchez, J. P.;

- Lopez Berrizbeitia, M. F.; Krasnov, B. R. (2024). Searching for common patterns in parasite ecology: species and host contributions to beta-diversity in helminths of South African ungulates and fleas of South American rodents. *International Journal for Parasitology* 54, 429-439.
- Ismayilova, R. I., Nagiyeva, B. A., Sadykhova, S. A., Rasulzade, Z. I., Amirova, K. R., Rustamova, M. K., & Gajialiyeva, G. G. (2024). Ecological and faunistic features of fleas in the Transcaucasian mounthill natural plague foci in Azerbaijan. *Parazitologîâ*, 58(4), 267-276.
- Jurado, V., Martin-Pozas, T., Fernandez-Cortes, A., Calaforra, J. M., Sanchez-Moral, S., & Saiz-Jimenez, C. (2024). Gypsum Cave Biofilm Communities are Strongly Influenced by Bat-And Arthropod-Related Fungi. *Microbial Ecology*, 87(1), 80.
- Košša, J., Ambros, M., Dudich, A., & Baláž, I. (2024). Impact of environmental factors on small mammals and their fleas. *Folia Oecologica*, 51(2), 146-153.
- Krasnov, B. R., Barki, G., & Khokhlova, I. S. (2024). Dissimilarity in flea and host assemblages and their interaction networks along a spatial distance gradient: different patterns revealed by different network dissimilarity metrics. *Oecologia*, 1-13.
- Krasnov, B. R., Khokhlova, I. S., Berrizbeitia, M. F. L., Matthee, S., Sanchez, J. P., & van der Mescht, L. (2024). Functional similarity affects similarity in partner composition in flea-mammal networks. *Parasitology Research*, 123(5), 203. doi.org/10.1007/s00436-024-08229-7.
- Krasnov, B., Khokhlova, I. S., Korrallo-Vinarskaya, N. P.; A. Laudisoit; López Berrizbietia, M. F.; Matthee, S., Sanchez, J. P., Stanko, M., van der Mescht, L.; M. V. Vinarski. (2024). Congruence between co-occurrence and trait-based networks is scale-dependent: a case study with flea parasites of small mammalian hosts. *Parasitology*, 1-11.

We applied a novel framework based on network theory and a concept of modularity that estimates congruence between trait-based (= functional) co-occurrence networks, thus allowing the inference of co-occurrence patterns and the determination of the predominant mechanism of community assembly. The aim was to investigate the relationships between species co-occurrence and trait similarity in flea communities at various scales (compound communities: across regions within a biogeographic realm or across sampling sites within a geographic region; component communities: across sampling sites within a geographic region; and infracommunities: within a sampling site). We found that compound communities within biogeographic realms were assembled via environmental or host-associated filtering. In contrast, functional and spatial/host-associated co-occurrence networks, at the scale of regional compound communities, mostly indicated either stochastic processes or the lack of dominance of any deterministic process. Analyses of congruence between functional and either spatial (for component communities) or host-associated (for infracommunities) co-occurrence networks demonstrated that assembly rules in these communities varied among host species. In component communities, stochastic processes prevailed, whereas environmental filtering was indicated in 4 and limiting similarity/competition in 9 of 31 communities. Limiting similarity/competition processes dominated in infracommunities, followed by stochastic mechanisms. We conclude that assembly processes in parasite communities are scale-dependent, with different mechanisms acting at different

scales.

Krasnov, B.R.; V. I. Grabovsky; I. S. Khokhlova; N. P. Korallo-Vinarskaya; M. F. López Berrizbietia; S. Matthee; J. P. Sanchez; M. Stanko; L. Van Der Mescht; M. V. Vinarski. (2024). Structure of compound and component communities of fleas parasitic on small mammals in six different regions as revealed by environmental based co-occurrence geometry analyses. *Integrative Zoology*.00, 1–15.

<https://doi.org/10.1111/1749-4877.12856>

Krasnov, B. R., Shenbrot, G. I., & Khokhlova, I. S. (2024). Nestedness of flea assemblages harboured by small mammalian hosts revisited: phylogenetic and functional nestedness do not follow compositional nestedness. *Parasitology Research*, 123(1), 111.

Krasnov, B. R., Shenbrot, G. I., Khokhlova, I. S., Lopez Berrizbeitia, M. F., Matthee, S., Sanchez, J. P., & Van der Mescht, L. (2024). Environment and traits affect parasite and host species positions but not roles in flea–mammal networks. *Integrative Zoology*.23 January 2024 <https://doi.org/10.1111/1749-4877.12799>

Krasnov, B. R., Khokhlova, I. S., Berrizbeitia, M. F. L., Matthee, S., Sanchez, J. P., Shenbrot, G. I., & van der Mescht, L. (2024). Relationships between functional alpha and beta diversities of flea parasites and their small mammalian hosts. *Parasitology*, 151(4), 449-460. <https://doi.org/10.1017/S0031182024000283>.

Li, K., Wang, Z., Wei, X., Ji, H., Shang, M., Chang, N., ... & Liu, Q. (2024). Relationships Between Meteorological Factors and Mongolian Gerbils and Its Flea Burdens—Xilingol League, Inner Mongolia Autonomous Region, China, 2012–2021. *China CDC Weekly*, 6(23), 547-552.

Майканов, Н. С., Танитовский, В. А., Жумадилова, З. Б., Садовская, В. П., & Просветова, И. К. (2024). МАТЕРИАЛЫ ПО ФАУНЕ БЛОХ (SIPHONAPTERA) ТЕПЛОКРОВНЫХ ЖИВОТНЫХ И ЖИЛЬЯ ЧЕЛОВЕКА ЗАПАДНОГО КАЗАХСТАНА. Степи Северной Евразии: материалы X международного симпозиума, (X), 808-813. [MATERIALS ON THE FAUNA OF FLEAS (SIPHONAPTERA) OF WARM-BLOODED ANIMALS AND HUMAN DWELLINGS IN WESTERN KAZAKHSTAN]

Padilla, C. J., Martin, J. T., Cain III, J. W., & Gompper, M. E. (2024). Abiotic and Demographic Drivers of Flea Parasitism on Deer Mice in a Recovering Mixed-Conifer Forest a Decade Postfire. *The Journal of Parasitology*, 110(4), 375-385.

With the intensity and frequency of wildfires increasing rapidly, the need to study the ecological effects of these wildfires is also growing. An understudied aspect of fire ecology is the effect fires have on parasite–host interactions, including ectoparasites that might be pathogen vectors. Although some studies have examined the impacts of fire on ticks, studies on other ectoparasites, including pathogen vectors, are rare. To help address this knowledge gap, we examined the abiotic and biotic factors that predict the likelihood and extent of parasitism of deer mice (*Peromyscus maniculatus*) by fleas within a landscape of unburned and recovering burned (>9 yr postfire) mixed conifer forests. We sampled 227 individual deer mice across 27 sites within the Jemez Mountains of northern New Mexico in 2022 and quantified measures of parasitism by fleas (primarily *Aetheca wagneri*). These sites were distributed in

both unburned areas ($n = 15$) and recovering burned areas ($n = 12$), with the latter derived from 2 large fires, the Las Conchas fire (2011) and the Thompson Ridge fire (2013). Using these data, we tested for differences in prevalence, mean abundance, and mean intensity of fleas on deer mice, focusing on the predictive importance of host sex and fire history. We also created generalized linear mixed-effects models to investigate the best host and environmental predictors of parasitism by fleas. Approximately a decade postfire, we found minimal evidence to suggest that fire history influenced either the presence or intensity of fleas on deer mice. Rather, at the current forest-regeneration stage, the extent of parasitism by fleas was best predicted by measures of host sex, body condition, and the trapline's ability to accumulate water, as measured through topography. As host body condition increased, the probability of males being parasitized increased, whereas the opposite pattern was seen for females. Male mice also had significantly greater flea loads. Among potential abiotic predictors, the topographic wetness index or compound topographic index (a proxy for soil moisture) was positively related to flea intensity, suggesting larger flea populations in burrows with higher relative humidity. In summary, although fire may potentially have short-term impacts on the likelihood and extent of host parasitism by fleas, in this recovering study system, host characteristics and topographic wetness index are the primary predictors of parasitism by fleas.

Benedek, A. M., Boeraş, I., Lazăr, A., Sandu, A., Cocîrlea, M. D., Stănciugelu, M., ... & Postolache, C. (2024). Effects of Season, Habitat, and Host Characteristics on Ectoparasites of Wild Rodents in a Mosaic Rural Landscape. *Animals*, 14(2), 304.

Rabitsch, W., Werenkraut, V., Kenis, M., Roques, A., & Roy, H. E. (2024). Invasive alien insects and insect decline. In *Biological Invasions and Global Insect Decline* (pp. 143-171). Academic Press.

Raimkulov, K., Fedorova, S., Sharsheeva, B., Kadyrova, B., & Dzhantaeva, G. (2024). The role of parasites in ecological diversity and its dynamics in the Kyrgyz Republic. In *E3S Web of Conferences* (Vol. 537, p. 05023). EDP Sciences.

This article examines the role of parasites in ecological diversity and their impact on ecosystem dynamics in the Kyrgyz Republic (KR). The research shows that parasites play an important role in maintaining the balance in nature and influence the species diversity of the region. Data on the interaction of parasites with animals and their impact on ecosystem health are presented. The analysis allows us to understand the mechanisms of interaction between parasites and the environment and propose measures to maintain ecological balance in the region. An analysis of the epidemiological situation of parasitosis in the Kyrgyz Republic for the period from 2008 to 2022 was also carried out. There is a trend of increasing incidence of diseases, with the nine most common nosological forms including enterobiasis, giardiasis, ascariasis, hymenolepiasis, teniarinosis, teniasis, dicroceliasis, fascioliasis, and trichocephaliasis. The average long-term incidence rate was 34.5 per 1000 individuals examined, with the minimum recorded in 2008 (32.6) and the maximum in 2012 (38.4).

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Stierhoff, E. N., Carpenetti, J. M., Knutie, S. A., Wallace, J. W., & Butler, M. W. (2024). Hematophagous ectoparasites affect nestling size and physiology, but not apparent survival. *Journal of Ornithology*, 165(2), 449-459.

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Zurita Carrasco, A., Trujillo, I., García Sánchez, Á. M., & Cutillas Barrios, C. (2024). Survey of flea infestation in cats and dogs in Western Andalusia, Spain: Seasonality and other risk factors for flea infestation. *Medical and Veterinary Entomology*, 38 (2), 244-251.

Zellner, P.N. & Brown, L.D. (2024) Influence of egg density on larval development and adult body size of cat fleas (*Ctenocephalides felis*). *Medical and Veterinary Entomology*, 1–6. Available from: <https://doi.org/10.1111/mve.12735>

Fleas (Siphonaptera) are holometabolous insects with larval and adult stages that exhibit vastly different ecologies from each other. Adult fleas are parasitic and feed exclusively on the blood of a vertebrate host, whereas flea larvae do not live on hosts and consume dried faecal blood from adult fleas. Because flea larvae rely on adult flea faeces for food, excrement and eggs must fall in the same location; thus, larval density is likely high in these restricted habitats. However, the influence of larval density on the subsequent adult stage has not been examined. In the present study, we utilized egg density to investigate density-dependent effects on larval development and adult body size in the cat flea (*Ctenocephalides felis* Bouché) (Siphonaptera: Pulicidae). Specifically, eggs were collected to create three different larval densities ($n = 50, 100$ and 150 per 56.7 cm^2), and hatched larvae from all groups were fed an excess amount of adult faecal pellets. Larval development was measured by recording the proportion of eggs that developed to the pupal stage and the proportion of eggs that reached adulthood (eclosion). The body size of eclosed adults was quantified for both sexes using head length and length of the total body. We found that the number of eggs had no effect on the proportion of larvae that pupated or the proportion of larvae that eclosed; however, higher egg densities resulted in larger body sizes for both sexes. Overall, these data yield significant insight into how the ecology of larval fleas impacts the biology of the resultant adults.

Zhang, L., Wang, Z., Chang, N., Shang, M., Wei, X., Li, K., ... & Liu, Q. (2024). Relationship between climatic factors and the flea index of two plague hosts in Xilingol League, Inner Mongolia Autonomous Region. *Biosafety and Health*, 6(4), 244-250.

Pathology and Control

Alfaia, F., Jota Baptista, C., Sós-Koroknai, V., Hoitsy, M., Sós, E., & Madeira de Carvalho, L. M. (2024). Hedgehogs' Parasitology: An Updated Review on Diagnostic Methods and Treatment. *Parasitologia*, 4(1), 82-90.

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Argueta, K., & Joshi, S. (2024). The Mystery of Seresto Collars. Celebration of Student Scholarship - Poster Presentations. 10. https://scholarworks.moreheadstate.edu/celebration_posters_2024/10

Elanco Animal Health's (EAH) Seresto collars are used to protect canines and felines from common Ixodids (ticks) and Siphonaptera (fleas). They work by slowly releasing the active ingredients onto the pet's skin throughout eight months. Recently, the Environmental Protection Agency (EPA) has revealed there have been a total of 100,000 incidents, 2,700 deaths in canines and felines, and 894 cases of harm to humans due to the use of Seresto collars. However, the link between these incidents and Seresto collars remains unclear. This literature review investigates the potential harm of Seresto collars. It compares the effects of major active ingredients, Flumethrin (a pyrethroid) and Imidacloprid (a neonicotinoid), to existing data on their potential toxicity in pets and humans. The study will analyze published research and surveys to assess if reported health effects (skin irritation, neurological problems, lethargy, seizures, death) are linked to Seresto collars or caused by counterfeit products.

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Brennan, R. N., Paulson, S. L., & Escobar, L. E. (2024). Estimating pathogen-spillover risk using host–ectoparasite interactions. *Ecology and Evolution*, 14(6), e11509.

Pathogen spillover corresponds to the transmission of a pathogen or parasite from an original host species to a novel host species, preluding disease emergence. Understanding the interacting factors that lead to pathogen transmission in a zoonotic cycle could help identify novel hosts of pathogens and the patterns that lead to disease emergence. We hypothesize that ecological and biogeographic factors drive host encounters, infection susceptibility, and cross-species spillover transmission. Using a rodent–ectoparasite system in the Neotropics, with shared ectoparasite associations as a proxy for ecological interaction between rodent species, we assessed relationships between rodents using geographic range, phylogenetic relatedness, and ectoparasite associations to determine the roles of generalist and specialist hosts in the transmission cycle of hantavirus. A total of 50 rodent species were ranked on their centrality in a network model based on ectoparasites sharing. Geographic proximity and phylogenetic relatedness were predictors for rodents to share ectoparasite species and were associated with shorter network path distance between rodents through shared ectoparasites. The rodent–ectoparasite network model successfully predicted independent data of seven known hantavirus hosts. The model predicted five novel rodent species as potential, unrecognized hantavirus hosts in South

America. Findings suggest that ectoparasite data, geographic range, and phylogenetic relatedness of wildlife species could help predict novel hosts susceptible to infection and possible transmission of zoonotic pathogens. Hantavirus is a high-consequence zoonotic pathogen with documented animal-to-animal, animal-to-human, and human-to-human transmission. Predictions of new rodent hosts can guide active epidemiological surveillance in specific areas and wildlife species to mitigate hantavirus spillover transmission risk from rodents to humans. This study supports the idea that ectoparasite relationships among rodents are a proxy of host species interactions and can inform transmission cycles of diverse pathogens circulating in wildlife disease systems, including wildlife viruses with epidemic potential, such as hantavirus.

Excerpt: "For example, in our rodent–ectoparasite network, the flea *Plocopsylla crypta*, had a strong relationship with geographic proximity of hosts, but a weak relationship with phylogenetic relatedness. This indicates that *P. crypta* is more of a specialist in terms of parasitizing hosts in a restricted geographic area, but within that area it can act as a generalist by parasitizing a wide range of distantly related host species."

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- The sand flea, *Tunga penetrans*, is the cause of a severely neglected parasitic skin disease (tungiasis) in the tropics and has received little attention from entomologists to understand its transmission ecology. Like all fleas, *T. penetrans* has environmental off-host stages presenting a constant source of reinfection. We adapted the Berlese-Tullgren funnel method using heat from light bulbs to extract off-host stages from soil samples to identify the major development sites within rural households in Kenya and Uganda.
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Naz, S., Saeed, R. F., Rajput, M., Abbasi, S. W., & Daniel, I. (2024). 4 Insecticide Resistance. In *Antiparasitic Drug Resistance in Veterinary Practice* (pp. 58-77). GB: CABI.

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Pauling, C. D., Beerntsen, B. T., Song, Q., & Anderson, D. M. (2024). Transovarial transmission of *Yersinia pestis* in its flea vector *Xenopsylla cheopis*. *Nature Communications*, 15(1), 7266.

Yersinia pestis, the causative agent of plague, is endemic in certain regions due to a stable transmission cycle between rodents and their associated fleas. In addition, fleas are believed to serve as reservoirs that can occasionally cause enzootic plague cycles and explosive epizootic outbreaks that increase human exposure. However, transmission by fleas is inefficient and associated with a shortened lifespan of the flea and rodent hosts, indicating that there remain significant gaps in our understanding of the vector-animal cycle of *Y. pestis*. Here, we show that laboratory-reared, infected fleas (*Xenopsylla cheopis*) can transmit viable *Y. pestis* from adults to eggs, and the bacteria can be passed through all subsequent life stages of the flea. Thus, our data raise the possibility that transovarial transmission in fleas might contribute to the persistence of *Y. pestis* in the environment without detectable plague activity in mammals.

Pawęska, J. T., Storm, N., Jansen van Vuren, P., Markotter, W., & Kemp, A. (2024). Attempted Transmission of Marburg Virus by Bat-Associated Fleas *Thaumapsylla breviceps breviceps* (Ichnopsyllidae: Thaumapsyllinae) to the Egyptian Rousette Bat (*Rousettus aegyptiacus*). *Viruses*, 16(8), 1197.

Egyptian rousette bats (ERBs) are implicated as reservoir hosts for Marburg virus (MARV), but natural mechanisms involved in maintenance of MARV in ERB populations remain undefined. A number of hematophagous ectoparasites, including fleas, parasitize bats. Subcutaneous (SC) inoculation of ERBs with MARV consistently results in viremia, suggesting that infectious MARV could be ingested by blood-sucking ectoparasites during feeding. In our study, MARV RNA was detected in fleas that took a blood meal during feeding on viremic bats on days 3, 7, and 11 after SC inoculation. Virus concentration in individual ectoparasites was consistent with detectable levels of viremia in the blood of infected host bats. There was neither seroconversion nor viremia in control bats kept in close contact with MARV-infected bats infested with fleas for up to 40 days post-exposure. In fleas inoculated intracoelomically, MARV was detected up to 14 days after intracoelomic (IC) inoculation, but the virus concentration was lower than that delivered in the inoculum. All bats that had

been infested with inoculated, viremic fleas remained virologically and serologically negative up to 38 days after infestation. Of 493 fleas collected from a wild ERB colony in Matlapitsi Cave, South Africa, where the enzootic transmission of MARV occurs, all tested negative for MARV RNA. While our findings seem to demonstrate that bat fleas lack vectorial capacity to transmit MARV biologically, their role in mechanical transmission should not be discounted. Regular blood-feeds, intra- and interhost mobility, direct feeding on blood vessels resulting in venous damage, and roosting behaviour of ERBs provide a potential physical bridge for MARV dissemination in densely populated cave-dwelling bats by fleas. The virus transfer might take place through inoculation of skin, mucosal membranes, and wounds when contaminated fleas are squashed during auto- and allogrooming, eating, biting, or fighting.

Keywords: Marburg virus; bat fleas; Egyptian rousette bat; transmission; vectorial capacity

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free-living and parasitic arthropods: evolutionary taxonomy. *Trends in Parasitology* 40, 8p744-762.

Highlights: An updated phylogenomic tree of the Nudiviridae is presented, including 49 species, proposing two genera associated with ectoparasitic insects.

Evolution of the family Nudiviridae is estimated to have occurred in the mid-Devonian, matching the timeline of the insect's evolution.

All the crustacean-infecting nudiviruses have likely emerged from a single common ancestor (~143 million years ago), perhaps transmitted to Jurassic crustaceans by an insect host.

The Nudiviridae have a proclivity for ectoparasitic arthropods (i.e., lice and fleas), linked to multiple diversification events in the Triassic (early Betanudivirinae split), Jurassic (involving ectoparasitic hosts of alphanudiviruses), and Cretaceous periods (Zeta/Etanudivirus diversification).

Nudiviruses inhabit various geographical and climatic zones, including members from subpolar habitats.

Abstract: The nudiviruses (family: Nudiviridae) are large double-stranded DNA (dsDNA) viruses that infect insects and crustaceans, and have most recently been identified from ectoparasitic members (fleas and lice). This virus family was created in 2014 and has since been expanded via the discovery of multiple novel viral candidates or accepted members, sparking the need for a new taxonomic and evolutionary overview. Using current information (including data from public databases), we construct a new comprehensive phylogeny, encompassing 49 different nudiviruses. We use this novel phylogeny to propose a new taxonomic structure of the Nudiviridae by suggesting two new viral genera (Zetanudivirus and Etanudivirus), from ectoparasitic lice. We detail novel emerging relationships between nudiviruses and their hosts, considering their evolutionary history and ecological role.

Keywords Arthropoda ectoparasite evolution virology paleovirology Nudiviridae

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Theerthalingam, N. (2024). Detection and comparative study of *Bartonella* pathogens in Lithuanian bat

fleas, Biologija / Biology (N010), Degree Discipline Magistro studijų programa / Magister study programme::Molekulinė biologija ir biotechnologijos / Molecular Biology and Biotechnology (M), Molekulinė biologija / Molecular Biology (D04) master thesis. Gamtos mokslų fakultetas / Faculty of Natural Sciences, Vytauto Didžiojo universitetas / Vytautas Magnus University, Biologijos katedra / Department of Biology.

Bat fleas, tiny blood-sucking insects (Siphonaptera: Ischnopsyllidae), are found globally except Antarctica and tend to prefer specific hosts and their potential role in carrying diseases. Identifying these flea species and the pathogens they might transmit is crucial for animal and human health. It also helps us understand flea behavior, ecology, and spread across different regions. This research focused on bat fleas from four districts of Lithuania. By examining them using molecular methods and genetic analysis, we aimed to determine the flea species and screen them for *Bartonella* pathogens they might be carrying. Nearly half (49%) of the examined flea samples (21 out of 43) tested positive for *Bartonella* spp. Further analysis revealed four flea species carried *Bartonella*, with potential genetic diversity among these bacterial strains. Interestingly, the detected *Bartonella* clustered with known *Bartonella* spp. in a genetic database, showing a high degree of similarity (98–99%) to *Bartonella* found in various hosts and their ectoparasites elsewhere (dissertation).

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Excerpt: "The biting flea, *Parapsyllus longicornis*, (Siphonaptera: Rhopalopsyllidae), is hypothesize[d] to be the most likely vector of APV for the Magellanic penguin... "

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Fleas in Art and History

Deans, A. R., & Porturas, L. (2024). Diversity and complexity of arthropod references in haiku. *Plos one*, 19(4), e0298865. <https://doi.org/10.31219/osf.io/mnkgb>

Haiku are short poems, each composed of about 10 words, that typically describe moments in nature. People have written haiku since at least the 17th century, and the medium continues to be popular with poets, amateurs, educators, and students. Collectively, these poems represent an opportunity to understand which aspects of nature—e.g., which taxa and biological traits—resonate with humans and whether there are temporal trends in their representation or the emotions associated with these moments. We tested this potential using a mix of linguistic and biological methods, in analyses of nearly 4,000 haiku that reference arthropods. We documented the taxa and the life history traits represented in these poems and how they changed over time. We also analyzed the poems for emotion and tone. Our results reveal a mix of predictable trends and compelling surprises, each of

which stand to potentially inform engagement strategies. At least 99 families of arthropods, in 28 orders, are represented in these haiku. The eight most commonly referenced taxa, from highest to lowest number of references, include: Lepidoptera, Hymenoptera, Diptera, Coleoptera, Araneae, Orthoptera, Hemiptera, and Odonata. Several common, conspicuous orders were never referenced, including Trichoptera, Plecoptera, and Megaloptera. The most commonly referenced traits relate to ecology (especially habitat, phenology, time of day), behavior (especially sound production), phenotype (especially color), and locomotion (especially flight). The least common traits in haiku relate to arthropod reproduction and physiology. Our analyses revealed few obvious temporal trends in the representations of taxa, biological traits, or emotion and tone. The broader implications of these results and possible future directions are discussed.

Excerpts of some letters concerning fleas by Anthonie van Leeuwenhoek 1632-1723

Fleas were important subjects to Leeuwenhoek as evidenced by the numerous letters he wrote describing them (Flea News 93). Translated from the Dutch.

Here are a few examples from: <https://lensonleeuwenhoek.net/>
<https://lensonleeuwenhoek.net/content/wrote-letter-67-of-1678-01-14-robert-hooke>

Excerpt Letter No. 37 [23]. January 14th 1678. Addressed to: R. Hooke, Secretary of the Royal Society.

Ick heb inde na somer vloij-wormen opgeslooten, om datse naar voetnoot haer omspinnen soudon, en eenige van dese spinsels, heb ick sedert weijnige dagen, ontstucken gebrooken, als wanneer uijt ijder vande selve, een vloij quam springen, Dogh deselve omme de koude waren seer traegh, en vielen veeltijts op sij, dit siende stelde ick vast, dat de Vloijen die int na jaer niet uijt comen, een gantsche winter, in haer gespin blijven leggen, en wanneer maer warmte vernemen, alsdan uijt het gespin dat sij gemaect hebben, (doen sij nogh wormen waren) comen.

[In some cases I have locked up flea worms, so that they would be spun around them, and some of these threads, I have recently discovered, have become loose, as if a flea were to jump out of their own hands, leaving them behind because of the cold, were very slow, and often fell on them. When I saw this, I noticed that the fleas, which do not come out after a year, continue to lay in their webs for the entire winter, and when they detect warmth, they then fall out of the cocoons that they made have, (if they were still worms) come.]

Excerpt Letter no. 127, 27th October, 1693, Addressed to: Petrus Rabus.

Het was mijn meening niet geweest, de Vloo-teelt met den druk gemeen te maken, ten ware, door uwe aanmoediging. Hier nevens gaat een afdruksel van 't gene ik daar omtrent aanmerkenswaardig hebbe nagespoort. Zoo ras en zullen mijne ontdekkingen niet gedrukt zijn, of ik zal de vryheid nemen, van UE die te laten toekomen.

...

De Vloo-teelt afgedrukt zijnde, zal ik onder de Pers laten komen de voortteling van de zoo

genaamde worm de Hemelt, mitsgaders die van het kleine veragte dierke de Mijt, welkers verzameling ons niet vreemder zal voorkomen, als die van de Vloo. enz.

[It had not been my intention to make Flea farming common practice, except for your encouragement. Attached here is a print of what I have noticed that was significant about it. No sooner will my discoveries be printed than I will take the liberty of letting you have them.

...

Having printed out the breeding of Fleas, I will bring under the Press the reproduction of the so-called worm the Hemelt, together with that of the small animal the Mite, whose collection will not seem to us any more strange than that of the Flea. etc.]

Excerpt Letter L-552 of 1717-06-15 to the members of the Royal Society:

De vleeschfibertjens van eenen Os niet meer dan 4 maal dikker, als die van een Vlooy. Hoe veel tyds dat 'er noodig is, op dat de Eyeren van Vlooyen volwassene Vlooyen werden, enz. Bevonden dat de Vlooyen een gansche winter, zonder voedsel, in haar gespin kunnen leeven.

[The meat fibers of an Ox are not more than 4 times thicker than those of a flea. How much time is needed for the eggs of fleas to become adult flea, etc. It was found that the fleas could live in their cocoons for an entire winter, without food.]

On the next page

De Geograaf [The Geographer], 1669

(Model thought to be Anthonie van Leeuwenhoek. 1632-1723)

by Johannes Vermeer

(baptized 31 October 1632, Delft, County of Holland, Dutch Republic [Netherlands]– buried 15 December 1675, Delft)

(oil on canvas, height: 53 cm (20.8 in) x width: 46.6 cm (18.3 in)

